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Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
GeS	*B*16	10.449(9); 3.653(4); 4.306(4)	
(Fe, Ge)Cu_2S_4	$E1_1$	5.332(1); 10.531(1)	
SnS_2	*C*6	3.66; 5.96	
SnS	*B*16	11.32(1); 4.050(4); 4.242(4) (several other structures also reported)	
$SnCu_2FeS_4$	tetr.	5.449(2); 10.757(3)	
PbS	*B*1	5.914 3(4) (Galena)	
$PbSnS_2$	*B*16	11.35; 4.05; 4.29 (Teallite)	
AsS	*B*1	7.153(4); 9.994(6); 12.986(6); 120.6° (Realgar)	
As_2S_3	*D*5*f*	4.256(2); 9.577(4); 12.191(5); 109.76(8)° (Orpiment)	
CoAsS	*C*2	5.576(2) (Cobaltite)	
NiAsS	$F0_1$	5.690 (Gersdorffite)	
CuAsS	ortho.	11.35; 5.456; 3.749	
Sb_2S_3	$D5_8$	11.380(5); 3.829(2); 11.189(5) (Stibnite)	
FeSbS	monocl.	6.02; 5.93; 6.02; 67.87° (Gudmundite)	
NiSbS	$F0_1$	5.933(3) (Ullmannite)	
$CuSbS_2$	$F5_6$	6.008(10); 3.784(10); 14.456(30) (Wolfsbergite)	
Bi_2S_3	$D5_8$	11.31; 3.98; 11.13 (Bismuthite)	
$NaBiS_2$	*B*1	5.75(2)	
$KBiS_2$	*B*1	6.00(1)	
$CuBiS_2$	$F5_6$	6.142 6(3); 3.918 9(4); 14.528 2(7) (Emplecite)	
$AgBiS_2$	*B*1	5.648	
	hex.	4.07(2); 19.06(5)	
Bi_2Te_2S	rhomb.	4.14; 29.45; 120°	

Table 6.2 STRUCTURAL DETAILS

See note at end of references for sources of these data.

*A*1 (Cu type)

Cubic: O_h^5–$Fm3m$; $a = 3.614\,91$; $A = 4$
Co-ordinates: 4Cu(O_h): 000, $\frac{1}{2}\frac{1}{2}0$ ↷

*A*2 (W type)

Cubic: O_h^9–$Im3m$; $a = 3.163\,79$; $A = 2$
Co-ordinates: 2W(O_h): 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

*A*3 (Mg type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 3.209\,44$, $c = 5.210\,76$; $A = 2$
Co-ordinates: 2Mg(D_{3h}): $\frac{2}{3}\frac{1}{3}0$; $\frac{1}{3}\frac{2}{3}\frac{1}{2}$

*A*4 (Diamond type)

Cubic: O_h^7–$Fd3m$; $a = 3.566986(3)$; $A = 8$
Co-ordinates: 8C(T_d): 000; $\frac{1}{2}\frac{1}{2}0$; ↷; $\frac{1}{4}\frac{1}{4}\frac{1}{4}$; $\frac{3}{4}\frac{3}{4}\frac{1}{4}$; ↷

*A*5 (Tin type)

Tetragonal: D_{4h}^{19}–$I4/amd$; $a = 5.830\,8$, $c = 3.181\,0$; $A = 4$
Co-ordinates: 4Sn(D_{2d}): 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$; $\frac{1}{2}0\frac{1}{4}$; $0\frac{1}{2}\frac{3}{4}$

*A*6 (In type)

Tetragonal: D_{4h}^{17}–$F4/mmm$; $a = 3.252\,2$, $c = 4.950\,7$; $A = 4$
Co-ordinates: 4In(D_{4h}): 000; $\frac{1}{2}\frac{1}{2}0$; ↷

A7 (As type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$
Co-ordinates:
Rhombohedral (I), 2As(C_{3v}): $\pm(xxx)$
Rhombohedral (II), 8As(C_{3v}): $(000; \frac{1}{2}\frac{1}{2}0;$ ↷$) \pm (xxx)$
Hexagonal (III), 6As(C_{3v}): $(000; \frac{2}{3}\frac{1}{3}\frac{1}{3}; \frac{1}{3}\frac{2}{3}\frac{2}{3}) \pm (00x)$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

	Rhomb. I $A=2$		*Rhomb II* $A=8$		*Hexagonal III* $A=6$			
	a	α	a	α	a	c	x	c/a
As	4.12	54° 10′	5.57	84° 38′	3.759 7(1)	10.441 2(2)	0.226	2.78
Sb	4.50	57° 06′	6.20	87° 24′	4.300 7	11.222	0.233	2.61
Bi	4.74	57° 14′	6.57	87° 32′	4.535 0(5)	11.814(1)	0.237	2.61
Simple cub.	—	60°	—	90°	—	—	—	2.45

*A*8 (Se type)

Hexagonal: D_3^4–$P3_121$ (and D_3^6–$P3_221$)
Co-ordinates: 3Se(C_2): $x00; \bar{x}\bar{x}\frac{1}{3}; 0x\frac{2}{3}$ (and $x00; \bar{x}\bar{x}\frac{2}{3}; 0x\frac{1}{3}$)

	a	c	c/a	x
Se	3.956(1)	5.069(2)	1.28	0.22
Te	4.312(1)	5.957(2)	1.38	0.27
Simple cub.	—	—	1.23	0.33

*A*9 (Graphite type)

(α) Hexagonal: D_{6h}^4–$P6_3/mmc$ (if $z=0$); or
C_{6v}^4–$P6_3/mc$ (if $z\neq 0$); $a=2.47(1), c=6.93(1); A=4$
Co-ordinates: 2C(D_{3h} or C_{3v}): $000; 00\frac{1}{2}$
2C(D_{3h} or C_{3v}): $\frac{1}{3}\frac{2}{3}z; \frac{2}{3}\frac{1}{3}(\frac{1}{2}+z); z\approx 0$ (or very probably $z=0$)
(β) Rhombohedral: D_{3d}^5–$R\bar{3}m; a=2.46, c=10.1; A=6$
Co-ordinates: 6C(C_{3v}): $(000; \frac{2}{3}\frac{1}{3}\frac{1}{3}; \frac{1}{3}\frac{2}{3}\frac{2}{3}) \pm (00x)$, with $x\approx\frac{1}{6}$ (or very probably $x=\frac{1}{6}$)

*A*10 (Hg type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$
Co-ordinates:
Rhombohedral (I): 1Hg (D_{3d}): 000
Rhombohedral (II): 2Hg (D_{3d}): $000; \frac{1}{2}\frac{1}{2}\frac{1}{2}$
Rhombohedral (III): 4Hg (D_{3d}): $000; \frac{1}{2}\frac{1}{2}0;$ ↻
Hexagonal (IV): 3Hg (D_{3d}): $000; \frac{1}{3}\frac{2}{3}\frac{1}{3}; \frac{2}{3}\frac{1}{3}\frac{2}{3}$

	Rhombohedral			*Hexagonal*	*Ideal cubic*			
	I	*II*	*III*	*IV*	*I*	*II*	*III*	*Hex.*
a	3.00	4.90	4.38	3.46	$\frac{a}{2}\sqrt{2}$	$\frac{a}{2}\sqrt{6}$	a	$\frac{a}{2}\sqrt{2}$
α	70° 32′	41° 25′	98° 15′	—	60°	33° 33′	90°	—
c	—	—	—	6.71	—	—	—	$a\sqrt{3}$
A	1	2	4	3	1	2	4	3
c/a	—	—	—	1.94	—	—	—	2.45

*A*11 (Ga type)

Orthorhombic: D_{2h}^{18}–$Abma; a=4.524\,2, b=4.519\,5, c=7.661\,8; A=8$
Co-ordinates: 8Ga(C_s): $(000; 0\frac{1}{2}\frac{1}{2}) \pm (x0z; \frac{1}{2}+x, \frac{1}{2}, z)$ with $x=0.079; z=0.153$

*A*12 (α-Mn type)

Cubic: T_d^3–$I\bar{4}3m; a=8.912\,5; A=58$
Co-ordinates: $(000; \frac{1}{2}\frac{1}{2}\frac{1}{2}) + 2\text{Mn}(T_d)$: 000
+8Mn(C_{3v}): $xxx; \bar{x}\bar{x}x;$ ↻; with $x=0.32$
+24Mn(C_s): $xxz;$ ↻; $\bar{x}\bar{x}z;$ ↻; $\bar{x}x\bar{z};$ ↻: $x\bar{x}\bar{z};$ ↻; with $x=0.36; z=0.04$
+24Mn(C_s) with similar co-ordinates but with $x=0.09; z=0.28$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

*A*13 (β-Mn type)

Cubic: O^6–$P4_132$ and O^7–$P4_132$; $a = 6.315(2)$; $A = 20$

Co-ordinates: 8Mn(C_3): $xxx; (\frac{1}{2}+x)(\frac{1}{2}-x)\bar{x}; \circlearrowleft$; $(\frac{3}{4}-x)(\frac{3}{4}-x)(\frac{3}{4}-x); (\frac{1}{4}-x)(\frac{3}{4}+x)(\frac{1}{4}+x); \circlearrowleft$; with $x = 0.061$

12Mn(C_2): $\frac{3}{8}\bar{x}(\frac{3}{4}+x); \circlearrowleft$; $\frac{7}{8}(\frac{1}{2}+x)(\frac{1}{4}-x); \circlearrowleft$; $\frac{1}{8}x(\frac{1}{4}+x); \circlearrowleft$; $\frac{5}{8}(\frac{1}{2}-x)(\frac{3}{4}-x); \circlearrowleft$; with $x = 0.206$

An alternative structure has been proposed (Wilson[9]) with space group O_h^6–$Fm3c$; $a = 12.58$; $A = 160$

*A*15 (Cr_3Si type)

Cubic: O_h^3–$Pm3n$; $a = 4.556(4)$; $A = 8$

Co-ordinates: 2Si(T_h): 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

6Cr(D_{2d}): $\frac{1}{2}0\frac{1}{4}; \circlearrowleft$; $\frac{1}{2}0\frac{3}{4}; \circlearrowleft$

*A*16 (orh. S type)

Orthorhombic: D_{2h}^{24}–$Fddd$; $a = 10.464\,6(1)$, $b = 12.866\,0(1)$, $c = 24.486\,0(3)$; $A = 128$

Co-ordinates: 4 times 32S(C_1) in $(000; \frac{1}{2}\frac{1}{2}0; \circlearrowleft) + xyz; \bar{x}y\bar{z}; (\frac{1}{4}-x)(\frac{1}{4}-y)(\frac{1}{4}-z); (\frac{1}{4}+x)(\frac{1}{4}-y)(\frac{1}{4}+z)$; $x\bar{y}\bar{z}; \bar{x}\bar{y}z; (\frac{1}{4}-x)(\frac{1}{4}+y)(\frac{1}{4}+z); (\frac{1}{4}+x)(\frac{1}{4}+y)(\frac{1}{4}-z)$

	x	*y*	*z*
S I	−0.017	0.083	0.072
S II	−0.094	0.161	0.200
S III	−0.167	0.105	0.125
S IV	−0.094	0.028	0.250

*A*20 (α-U type)

Orthorhombic: D_{2h}^{17}–$Cmcm$; $a = 2.844\,4$, $b = 5.868\,9$, $c = 4.931\,6$; $A = 4$

Co-ordinates: 4U(C_{2v}): $(000; \frac{1}{2}\frac{1}{2}0) + 0y\frac{1}{4}; 0\bar{y}\frac{3}{4}; y = 0, 105$

A_a (Pa type)

Tetragonal: D_{4h}^{17}–I^4/mmm; $a = 3.940$, $c = 3.244$; $A = 2$

Co-ordinates: 2Pa(D_{4h}): 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

A_b (β-U type)

Tetragonal: C_{4v}^4–$P4nm$; $a = 10.758\,9$, $c = 5.653\,1$; $A = 30$

Co-ordinates: 2U(C_{2v}): $00z; \frac{1}{2}\frac{1}{2}(\frac{1}{2}+z); z = 0.66$

4U(C_s): $xxz; \bar{x}\bar{x}z; (\frac{1}{2}+x)(\frac{1}{2}-x)(\frac{1}{2}+z); (\frac{1}{2}-x)(\frac{1}{2}+x)(\frac{1}{2}+z); x = 0.11; z = 0.23$

4U(C_s) in similar position with $x = 0.32; z = 0.00$

4U(C_s) in similar position with $x = 0.68; z = 0.50$

8U(C_1): $xyz; \bar{x}\bar{y}z; (\frac{1}{2}+x)(\frac{1}{2}-y)(\frac{1}{2}+z); (\frac{1}{2}-x)(\frac{1}{2}+y)(\frac{1}{2}+z); xyz; \bar{y}\bar{x}z; (\frac{1}{2}+y)$ $(\frac{1}{2}-x)(\frac{1}{2}+z); (\frac{1}{2}-y)(\frac{1}{2}+x)(\frac{1}{2}+z)$; with $x = 0.56; y = 0.24; z = 0.25$

8U(C_1) in similar position with $x = 0.38; y = 0.04; z = 0.20$

Thewlis[14] compared the lattice constants at 720°C of pure β-U and Cr containing β-U.

	a	*c*
Pure β-U at 720°C	10.759	5.656
1.4 at % Cr-U alloy at 720°C	10.763	5.652
1.4 at % Cr-U alloy at 20°C	10.590	5.634

A_c (α-Np)

Orthorhombic: D_{2h}^{16}–$Pmcn$; $a = 6.663(3)$, $b = 4.723(1)$, $c = 4.887(2)$; $A = 8$

Co-ordinates: 4Np(C_s): $\pm(\frac{1}{4}yz); \pm(\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}+z); y = 0.208; z = 0.036$

4Np(C_s) in similar positions with $y = 0.842; z = 0.319$

A_d (β-Np)

Tetragonal: D_4^2–$P42_1$; $a = 4.897(2)$, $c = 3.388(2)$; $A = 4$

Co-ordinates: 2Np(D_2): 000, $\frac{1}{2}\frac{1}{2}0$

2Np(C_4): $\frac{1}{2}0z; 0\frac{1}{2}\bar{z}; z = 0.38$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

A_e (β'-$TiCu_3$ type)

Orthorhombic: D_{2h}^{17}–$Cmcm$; $a = 2.585, b = 4.527, c = 4.351$; $A = 4$
Co-ordinates: 4Ti or Cu(C_{2v}): $(000; \frac{1}{2}\frac{1}{2}0) + 0y\frac{1}{4}; 0\bar{y}\frac{3}{4}; y = 0.345$

A_f ($HgSn_{6-13}$ type)

Hexagonal: D_{6h}^{1}–$P6/mmm$; $a = 3.205, c = 2.984$; $A = 1$ (data for $HgSn_6$)
Co-ordinates: 1Hg or Sn (D_{6h}): 000

A_g (B type)

Tetragonal: $P4_2/nnm$; $a = 8.75(2), c = 5.075(1)$; $A = 50$
Co-ordinates: 2B(S_4): $00\frac{1}{2}; \frac{1}{2}\frac{1}{2}0$
6 times 8B (C_1): $xyz; (\frac{1}{2}-x)(\frac{1}{2}+y)(\frac{1}{2}+z)$
$\bar{x}\bar{y}z; (\frac{1}{2}+x)(\frac{1}{2}-y)(\frac{1}{2}+z)$
$\bar{y}x\bar{z}; (\frac{1}{2}+y)(\frac{1}{2}+x)(\frac{1}{2}-z)$
$y\bar{x}\bar{z}; (\frac{1}{2}-y)(\frac{1}{2}-x)(\frac{1}{2}-z)$

	B I	*B II*	*B III*	*B IV*	*B V*	*B VI*
x	0.328	0.095	0.223	0.078	0.127	0.250
y	0.095	0.328	0.078	0.223	0.127	0.250
z	0.395	0.395	0.105	0.105	0.395	−0.078

A_h (α-Po type)

Cubic: O_h^1–$Pm3m$; $a = 3.359(1)$; $A = 1$
Co-ordinates: 1Po(O_h): 000

A_i (β-Po)

Rhombohedral: D_{3d}^5–$R\bar{3}m$; $a = 3.37$; $\alpha = 98°13'$; $A = 1$
Co-ordinates: 1Po(D_{3d}^5): 000

A_k (α-Se)

Monoclinic: C_{2h}^5–$P2_1/n$; $a = 9.31$; $b = 8.07$; $c = 12.85$; $\beta = 93.13°$; $A = 32$
Co-ordinates: 8 times 4Se(C_1): $\pm(xyz) \pm (\frac{1}{2}+x, \frac{1}{2}-y, \frac{1}{2}+z)$

	Se *I*	Se *II*	Se *III*	Se *IV*	Se *V*	Se *VI*	Se *VII*	Se *VIII*
x	0.321	0.427	0.317	0.134	−0.081	−0.156	−0.084	0.131
y	0.486	0.664	0.637	0.820	0.686	0.733	0.520	0.597
z	0.237	0.357	0.535	0.556	0.521	0.328	0.229	0.134

A_l (β-Se)

Monoclinic: C_{2h}^5–$P2_1/a$; $a = 12.85, b = 8.07, c = 9.31$; $\beta = 93°\ 08'$; $A = 32$
Co-ordinates: 8 times 4Se(C_1): $\pm(xyz) \pm (\frac{1}{2}+x, \frac{1}{2}-y, z)$

	Se *I*	Se *II*	Se *III*	Se *IV*	Se *V*	Se *VI*	Se *VII*	Se *VIII*
x	0.334	0.227	0.080	0.102	0.159	0.340	0.409	0.459
y	0.182	0.221	0.397	0.578	0.832	0.832	0.763	0.476
z	0.436	0.245	0.238	0.050	0.157	0.141	0.366	0.336

*B*1 (NaCl type)

Cubic: O_h^5–$Fm3m$; $a = 5.640$; $A = 8$
Co-ordinates: $(000; \frac{1}{2}\frac{1}{2}0; \text{♪}) + 4\text{Na}(O_h)$: 000
$+ 4\text{Cl}(O_h)$: $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

*B*2 (CsCl type)

Cubic: O_h^1–$Pm3m$; $a = 4.123$; $A = 2$
Co-ordinates: Cs(O_h): 000; Cl(O_h): $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

*B*3 [Sphalerite (ZnS) type]

Cubic: T_d^2–$F\bar{4}3m$; $a = 5.382\,9$; $A = 8$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ↻) + 4Zn(T_d): 000
+ 4S(T_d): $\frac{1}{4}\frac{1}{4}\frac{1}{4}$

*B*4 [Wurtzite(ZnS) type]

Hexagonal: C_{6v}^4–$P6_3mc$; $a = 3.822\,7(1)$, $c = 6.260\,7(1)$; $A = 4$
Co-ordinates: 2Zn(C_{3v}): $\frac{1}{3}\frac{2}{3}0$; $\frac{2}{3}\frac{1}{2}\frac{1}{2}$
2S(C_{3v}): $\frac{1}{3}\frac{2}{3}z$; $\frac{2}{3}\frac{1}{3}(\frac{1}{2}+z)$; $z \approx \frac{3}{8}$

*B*8 (α-NiAs type; β-Ni_2 In type)

Between the main types (α) and (β) there exist a number of intermediate arrangements due to the variation of the stoichiometric formulae. The axial ratio c/a may change from the value 1.75 (in type α) to 1.22 (in type β). Similarly, there is virtually a continuous change from the B8 type to the C6 type.

α-NiAs type ($B8_1$)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 3.619$, $c = 5.044$, $c/a = 1.39$; $A = 4$
Co-ordinates: 2Ni(D_{3d}): $000; 00\frac{1}{2}$
2As(D_{3h}): $\frac{1}{3}\frac{2}{3}\frac{1}{4}$; $\frac{2}{3}\frac{1}{3}\frac{3}{4}$

β-Ni_2 In type ($B8_2$)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 4.188\,9$, $c = 5.123\,0$, $c/a = 1.22$; $A = 6$
2Ni(D_{3d}): $000; 00\frac{1}{2}$; 2Ni(D_{3h}): $\frac{1}{3}\frac{2}{3}\frac{3}{4}$; $\frac{2}{3}\frac{1}{3}\frac{1}{4}$
2In(D_{3h}): $\frac{1}{3}\frac{2}{3}\frac{1}{4}$; $\frac{2}{3}\frac{1}{3}\frac{3}{4}$

*B*9 [Cinnabar (HgS) type]

Hexagonal: D_3^4–$P3_121$ and D_3^6–$P3_221$; $a = 4.147\,9(7)$, $c = 9.499\,60(15)$; $A = 6$
Co-ordinates: 3Hg(C_2): $x00; \bar{x}\bar{x}\frac{1}{3}; 0x\frac{2}{3}$; $x = 0.33$
3S(C_2): $x0\frac{1}{2}; \bar{x}\bar{x}\frac{5}{6}; 0x\frac{1}{6}$; $x = 0.21$

*B*10 (LiOH type)

Tetragonal: D_{4h}^7–$P4/nmm$; $a = 3.552$, $c = 4.347$; $A = 4$
Co-ordinates: 2Li(D_{2d}): $000; \frac{1}{2}\frac{1}{2}0$
2OH(C_{4v}): $0\frac{1}{2}z$; $\frac{1}{2}0\bar{z}$; $z = 0.20$
For FeSe: $z = 0.26$

*B*11 (PbO type)

Tetragonal: D_{4h}^7–$P4/nmm$; $a = 3.974\,4(5)$, $c = 5.022\,0(5)$; $A = 4$
Co-ordinates: 2Pb(C_{4v}): $0\frac{1}{2}z$; $\frac{1}{2}0\bar{z}$; $z = 0.24$
2O(C_{4v}): the same with $z = 0.74$
For γ-TiCu: $z(\text{Ti}) = 0.65$; $z(\text{Cu}) = 0.10$

*B*13 [Millerite (NiS) type]

Rhombohedral: C_{3v}^5–$R3m$; $a = 5.64$; $\alpha = 116°\,35'$; $A = 6$
Co-ordinates: 3Ni(C_s): xxz; ↻; $x = 0$; $z = 0.264$
3S(C_s): the same with $x = 0.714$; $z = 0.361$

*B*16 (GeS type)

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 4.306(4)$, $b = 10.449(9)$, $c = 3.653(4)$; $A = 8$
Co-ordinates: 4Ge(C_s): $\pm(xy\frac{1}{4})$; $\pm[(\frac{1}{2}-x)(\frac{1}{2}+x)\frac{1}{4}]$; $x = 0.167$; $y = 0.375$
4S(C_s): the same with $x = 0.1\bar{1}1$; $y = 0.139$

*B*17 [Cooperite (PtS) type]

Tetragonal: D_{4h}^9–$P4_2/mmc$; $a = 3.470\,1(3)$, $c = 6.109\,2(8)$; $A = 4$
Co-ordinates: 2Pt (D_{2h}): $0\frac{1}{2}0$; $\frac{1}{2}0\frac{1}{2}$, 2S(D_{2d}): $00\frac{1}{4}; 00\frac{3}{4}$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

*B*18 [Covellite (CuS) type]

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 3.768(3), c = 16.27(6); A = 12$
Co-ordinates: 2Cu(D_{3h}): $\pm(\frac{2}{3}\,\frac{1}{3}\,\frac{1}{4})$
4Cu(C_{3v}): $\pm(\frac{1}{3}\,\frac{2}{3}z); \pm(\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z); z = 0.107$
2S(D_{3h}): $\pm(\frac{1}{3}\,\frac{2}{3}\,\frac{1}{4})$
4S(C_{3v}): $\pm(00z); \pm(0, 0, \frac{1}{2} - z); z = 0.063$

*B*19 (AuCd type)

Orthorhombic: D_{2h}^5–$Pmcm$; $a = 3.143\,7, b = 4.871\,7, c = 4.769\,0; A = 4$
Co-ordinates: 2Au(C_{2v}): $\pm(0y\frac{1}{4}); y = 0.805$
2Cd(C_{2v}): $\pm(\frac{1}{2}y\frac{1}{4}); y = 0.315$
For MgCd: $y(\text{Mg}) = 0.818; y(\text{Cd}) = 0.323$

*B*20 (FeSi type)

Cubic: T^4–$P2_13$; $a = 4.483; A = 8$
Co-ordinates: 4Fe(C_3): $xxx; (\frac{1}{2} + x)(\frac{1}{2} - x)\bar{x}; \bar{x}; x = 0.137$
4Si(C_3): the same with $x = -0.158$
For BeAu: $x(\text{Be}) = -0.156; x(\text{Au}) = 0.150$
For RhSn: $x(\text{Rh}) = 0.142; x(\text{Sn}) = 0.159$

*B*27 (FeB type)

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 4.064, b = 5.503, c = 2.946; A = 8$
Co-ordinates: 4Fe(C_s): $\pm(xy\frac{1}{4}); \pm(\frac{1}{2} - x, \frac{1}{2} + y, \frac{1}{4}); x = 0.125; y = 0.180$
4B(C_s): the same with $x = 0.61; y = 0.036$
For MnB: $x(\text{Mn}) = 0.125; y(\text{Mn}) = 0.180; x(\text{B}) = 0.614; y(\text{B}) = 0.031$
For USi: $x(\text{U}) = 0.125; y(\text{U}) = 0.180; x(\text{Si}) = 0.611; y(\text{Si}) = 0.028$
For TiB: $x(\text{Ti}) = 0.123; y(\text{Ti}) = 0.177; x(\text{B}) = 0.603; y\text{B}) = 0.029$

*B*29 (SnS type)

Orthorhombic: D_{2h}^{16}–$Pmcn$; $a = 4.050(4), b = 4.242(4), c = 11.32(1); A = 8$
Co-ordinates: 4Sn(C_s): $\pm(\frac{1}{4}yz); \pm(\frac{1}{4}, \frac{1}{2} - y, \frac{1}{2} + z); y = 0.115; z = 0.118$
4S(C_s): the same with $y = 0.478; z = 0.850$
If this description, given in Strukturbericht vol. **3**, p. 14, is transformed to the following, it is virtually identical with the B16 (GeS type).
Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 4.33, b = 11.18, c = 3.98; A = 8$
Co-ordinates: 4Sn(C_s) $\pm(xy\frac{1}{4}); \pm(\frac{1}{2} - x, \frac{1}{2} + y, \frac{1}{4}); x = 0.115; y = 0.382$
4S(C_s): the same with $x = 0.022; y = 0.150$

*B*31 (MnP type)

Orthorhombic: D_{2h}^{16}–$Pcmn$; $a = 5.902\,7(4), b = 3.180\,6(2), c = 5.240\,9(3); A = 8$
Co-ordinates: 4Mn(C_s): $\pm(x\frac{1}{4}z); \pm(\frac{1}{2} - x, \frac{1}{4}, \frac{1}{2} + z); x = 0.20; z = 0.005$
4P(C_s): the same with $x = 0.57; z = 0.19$
For AuGa: $x(\text{Au}) = 0.184; z(\text{Au}) = 0.010; x(\text{Ga}) = 0.590; z(\text{Ga}) = 0.195$
For PdSi: $x(\text{Pd}) = 0.190; z(\text{Pd}) = 0.070; x(\text{Si}) = 0.570; z(\text{Si}) = 0.190$
For PtSi: $x(\text{Pt}) = 0.195; z(\text{Pt}) = 0.010; x(\text{Si}) = 0.590; z(\text{Si}) = 0.195$
For NiGe: $x(\text{Ni}) = 0.190; z(\text{Ni}) = 0.005; x(\text{Ge}) = 0.583; z(\text{Ge}) = 0.188$
For PdGe: $x(\text{Pd}) = 0.188; z(\text{Pd}) = 0.005; x(\text{Ge}) = 0.595; z(\text{Ge}) = 0.190$
For IrGe: $x(\text{Ir}) = 0.192; z(\text{Ir}) = 0.010; x(\text{Ge}) = 0.590; z(\text{Ge}) = 0.185$
For PtGe: $x(\text{Pt}) = 0.195; z(\text{Pt}) = 0.010; x(\text{Ge}) = 0.590; z(\text{Ge}) = 0.195$
For PdSn: $x(\text{Pd}) = 0.182; z(\text{Pd}) = 0.007; x(\text{Sn}) = 0.590; z(\text{Sn}) = 0.182$
For RhSb: $x(\text{Rh}) = 0.192; z(\text{Rh}) = 0.010; x(\text{Sb}) = 0.590; z(\text{Sb}) = 0.195$
For NiSi: $x(\text{Ni}) = 0.184; z(\text{Ni}) = 0.006; x(\text{Si}) = 0.580; z(\text{Si}) = 0.170$

*B*32 (NaTl type)

Cubic: O_h^7–$Fd3m$; $a = 7.480(4); A = 16$
Co-ordinates: $(000; \frac{1}{2}\,\frac{1}{2}\,0; \bar{x})$ + 8Na(T_d): $000; \frac{1}{4}\,\frac{1}{4}\,\frac{1}{4}$
+ 8Tl(T_d): $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}; \frac{3}{4}\,\frac{3}{4}\,\frac{3}{4}$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

*B*34 (PdS type)

Tetragonal: C_{4h}^2–$P4_2/m$; $a = 6.429(2), c = 6.611(2); A = 16$
Co-ordinates: 2Pd(S_4): $00\frac{1}{4}; 00\frac{3}{4}$; 2*Pd*($C_{2h}$): $0\frac{1}{2}0; \frac{1}{2}0\frac{1}{2}$
4Pd(C_s): $\pm(xy0); \pm(\bar{x}y\frac{1}{2})$; $x = 0.475; y = 0.250$
8S(C_1): $\pm(xyz); \pm(xy\bar{z}); \pm(y, \bar{x}, \frac{1}{2}+z); \pm(y, \bar{x}, \frac{1}{2}-z)$
with $x = 0.20; y = 0.32; z = 0.22$

*B*35 (CoSn type)

Hexagonal: D_{6h}^1–$P6/mmm$; $a = 5.278, c = 4.258; A = 6$
Co-ordinates: 1Sn(D_{6h}): 000; 2Sn(D_{3h}): $\frac{1}{3}\frac{2}{3}\frac{1}{2}; \frac{2}{3}\frac{1}{3}\frac{1}{2}$
3Co(D_{2h}): $\frac{1}{2}00; 0\frac{1}{2}0; \frac{1}{2}\frac{1}{2}0$

*B*37 (TlSe type)

Tetragonal: D_{4h}^{18}–$I4/mcm$; $a = 8.024, c = 7.18; A = 16$
Co-ordinates: $(000; \frac{1}{2}\frac{1}{2}\frac{1}{2})$ +4Tl(D_4): $00\frac{1}{4}; 00\frac{3}{4}$; +4Tl($D_{2d}$): $\frac{1}{2}0\frac{1}{4}; \frac{1}{2}0\frac{3}{4}$
+8Se(C_{2v}): $\pm(x, \frac{1}{2}+x, 0); \pm(\frac{1}{2}+x, \bar{x}, 0); x = 0.179$

B_a (UCo type)

Cubic: T^5–$I2_13$; $a = 6.352\,5; A = 16$
Co-ordinates: $(000; \frac{1}{2}\frac{1}{2}\frac{1}{2})$ + 8U(C_3): $xxx; (\frac{1}{2}+x)(\frac{1}{2}-x)\bar{x}; \circlearrowleft; x = 0.035$
+ 8Co(C_3): the same with $x = 0.294$

B_b (ζ-AgZn type)

Hexagonal: C_{3i}^1–$P\bar{3}$; $a = 7.636\,0; c = 2.819\,7; A = 9$
Co-ordinates: 1Zn(C_{3i}): 000; +2Zn(C_3): $\frac{1}{3}\frac{2}{3}z; \frac{2}{3}\frac{1}{3}z; z \approx \frac{3}{4}$
(1.5Zn + 4.5Ag) (C_1): $\pm(xyz)$; $\pm(\bar{y}, x-y, z); \pm(y-x, \bar{x}, z)$ with $x = 0.350$;
$y = 0.032; z = 0.750$

B_c (CaSi type)

Orthorhombic: D_{2h}^{17}–$Cmmc$; $a = 3.90(3), b = 4.545(3), c = 10.728(10); A = 8$
Co-ordinates: $(000; 0\frac{1}{2}\frac{1}{2})$ +4Ca(C_{2v}): $\pm(\frac{1}{4}0z); z = 0.36$
+4Si(C_{2v}): the same with $z = 0.07$
By choosing different axes and origin from those given in the original paper, this type becomes virtually identical with the *Bf* (CrB type):
D_{2h}^{17}–$Cmcm$; $a = 4.59, b = 10.80, c = 3.91$; y(Ca) = 0.14; y(Si) = 0.43

B_d (η-NiSi)

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 5.628(2), b = 5.190(1), c = 3.330(1); A = 8$
Co-ordinates: 4Ni(C_s): $xy0; \bar{x}\bar{y}\frac{1}{2}; (\frac{1}{2}-x)(\frac{1}{2}+y)0; (\frac{1}{2}+x)(\frac{1}{2}-y)\frac{1}{2}; x = 0.184; y = 0.006$
4Si(C_s): the same with $x = 0.080; y = 0.330$
By choosing different axes and origin from those given in the original paper, this type becomes identical with the *B*31(MnP type):
D_{2h}^{16}–$Pcmn$; $A = 5.62, b = 3.34, c = 5.18$; x(Ni) = 0.184; z(Ni) = 0.006; x(Si) = 0.580;
z(Si) = 0.170

B_e (CdSb type)

Orthorhombic: D_{2h}^{15}–$Pbca$; $a = 6.476\,9(1), b = 8.251(2), c = 8.522(2); A = 16$
Co-ordinates: 8Sb(C_1): $\pm(xyz); \pm(\frac{1}{2}+x, \frac{1}{2}-y, \bar{z}); \pm(\bar{x}, \frac{1}{2}+y, \frac{1}{2}-z); \pm(\frac{1}{2}-x, \bar{y}, \frac{1}{2}+z)$
8Cd(C_1): the same

	Sb			Cd *or* Zn		
	x	y	z	x	y	z
CdSb	0.136	0.072	0.108	0.456	0.119	−0.128
ZnSb	0.142	0.081	0.111	0.461	0.103	−0.122

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

B_f (CrB type)

Orthorhombic: D_{2h}^{17}–$Cmcm$; $a = 2.970, b = 7.865, c = 2.936; A = 8$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$) +4Cr(C_{2v}): $\pm(0y\frac{1}{4})$; $y = 0.146$
+4B(C_{2v}): the same with $y = 0.440$
For NbB: y(Nb) = 0.146; y(B) = 0.444
For CaSi: y(Ca) = 0.14; y(Si) = 0.43 (cf. Bc type)

B_g (MoB type)

Tetragonal: D_{4h}^{19}–$I4/amd$; $a = 3.103, c = 16.95; A = 16$
Co-ordinates: (000, $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 8Mo($C_{2v}$): $\pm(00z); \pm(0, \frac{1}{2}, \frac{1}{4} + z); z = 0.197$
+ 8B(C_{2v}): the same with $z = 0.35$

B_h (WC type)

Hexagonal: D_{6h}^{1}–$P\bar{6}m2 = 2.906, c = 2.837; A = 2$
Co-ordinates: 1W(D_{6h}): 000; 1C(D_{3h}): $\frac{1}{3}\frac{2}{3}\frac{1}{2}; \frac{2}{3}\frac{1}{3}\frac{1}{2}$

B_i (γ'-MoC type)

Hexagonal: D_{6h}^{4}–$P6_3/mmc$; $a = 2.932, c = 10.97; A = 8$
Co-ordinates: 4Mo(C_{3h}): $\frac{1}{3}\frac{2}{3}z; \frac{1}{3}\frac{2}{3}(\frac{1}{2} - z); \frac{2}{3}\frac{1}{3}(\frac{1}{2} + z); \frac{2}{3}\frac{1}{3}\bar{z}; z \approx \frac{1}{8}$
4C: in holes

B_l [Realgar (AsS) type]

Monoclinic: C_{2h}^{5}–$P2_1n$; $a = 9.994(6), b = 12.986(6), c = 7.153(4); \beta = 120.6°; A = 32$
Co-ordinates: 4 times 4As(C_1) and 4 times 4S(C_1) in:
$\pm(xyz); \pm(\frac{1}{2} + x, \frac{1}{2} - x, \frac{1}{2} + z)$

	As *I*	As *II*	As *III*	As *IV*	S *I*	S *II*	S *III*	S *IV*
x	0.118	0.425	0.318	0.038	0.346	0.213	0.245	0.115
y	0.024	−0.140	−0.127	−0.161	0.008	0.024	−0.225	−0.215
z	−0.241	−0.142	0.181	−0.290	−0.295	0.120	−0.363	0.048

B_m (TiB type)

Orthorhombic: D_{2h}^{16}–$Pnma$; $a = 6.105, b = 3.048, c = 4.551, A = 8$
Co-ordinates: 4Ti(C_s): $\pm(x\frac{1}{4}z); \pm(\frac{1}{2} - x, \frac{3}{4}, \frac{1}{2} + z); x = 0.177; z = 0.123$
4B(C_s): the same with $x = 0.029; z = 0.603$
If the axes are changed from those of the original paper, this type becomes identical with the $B27$ (FeB) type:

D_{2h}^{16}–$Pbnm$; $a = 4.56, b = 6.12, c = 3.06$; x(Ti) = 0.123; y(Ti) = 0.177;
x(B) = 0.603; y(B) = 0.029

$C1$ (CaF_2 type—MgAgAs type)

(α) Cubic: O_h^5–$Fm3m$; $a = 5.462\,6; A = 12$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; $\mathcal{J}$) +4Ca(O_h): 000
+8F(T_d): $\pm(\frac{1}{4}\frac{1}{4}\frac{1}{4})$
In those cases in which the F-position is occupied by two components in an ordered fashion—for example in As(MgAg)—the space group is changed to
(β) Cubic: T_d^2–$F\bar{4}3m$; $a = 6.24; A = 12$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; $\mathcal{J}$) +4As(T_d): 000
+ 4Ag(T_d): $\frac{1}{4}\frac{1}{4}\frac{1}{4}$
+ 4Mg(T_d): $\frac{3}{4}\frac{3}{4}\frac{3}{4}$

$C2$ [Pyrites (FeS_2) type]

Cubic: T_h^6–$Pa3$; $a = 5.293(2); A = 12$
Co-ordinates: 4Fe(S_{3i}): 000; $\frac{1}{2}\frac{1}{2}0$;$\mathcal{J}$
8S(C_3): $\pm(xxx); \pm(\frac{1}{2} + x, \frac{1}{2} - x, \bar{x}; \mathcal{J}); x = 0.386$
For MnS_2, $x = 0.401$; for CoAsS and NiAsS (random distribution of As and S) $x = 0.385$; for $PtBi_2$: $x = 0.38$.

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

*C*6 (CdI_2 type)

Hexagonal: D_{3d}^3–$P\bar{3}m1$; $a = 4.244$; $c = 6.859$; $A = 3$
Co-ordinates: 1Cd(D_{3d}): 000; 2I(c_{3v}): $\frac{1}{3}\,\frac{2}{3}z$; $\frac{2}{3}\,\frac{1}{3}\bar{z}$; $z \approx \frac{1}{4}$
There is virtually a continuous change from this type to the *B*8 type.

*C*7 (MoS_2 type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 3.159(2)$, $c = 12.307(2)$; $A = 6$
Co-ordinates: 2Mo(D_{3h}): $\frac{1}{3}\,\frac{2}{3}\,\frac{1}{4}$; $\frac{2}{3}\,\frac{1}{3}\,\frac{1}{4}$
4S(C_{3v}): $\frac{1}{3}\,\frac{2}{3}z$; $\frac{2}{3}\,\frac{1}{3}\bar{z}$; $\frac{1}{3}\,\frac{2}{3}(\frac{1}{2} - z)$; $\frac{2}{3}\,\frac{1}{3}(\frac{1}{2} + z)$; $z = 0.62$

*C*11*a* (CaC_2 type)

Tetragonal: D_{4h}^{17}–$I4/mmm$; $a = 3.89(1)$, $c = 6.38(1)$, $c/a = 1.64$; $A = 6$
Co-ordinates: (000; $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}$) +2Ca($D_{4h}$): 000
+4C(C_{4v}): $\pm(00z)$; $z = 0.38$

*C*11*b* ($MoSi_2$ type)

Tetragonal: D_{4h}^{17}–I/mmm; $a = 3.208$, $c = 7.900$, $c/a = 2.46$; $A = 6$
Co-ordinates: (000; $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}$ +2Mo(D_{4h}): 000
+4Si(C_{4v}): $\pm(00z)$; $z \approx \frac{1}{3}$
This type is a superstructure of the Aα(Pa) type.

*C*12 ($CaSi_2$ type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$; $a = 10.4$; $\alpha = 21°30'$; $A = 6$
Co-ordinates: 2Ca(C_{3v}): $\pm(xxx)$; $x = 0.083$
2Si(C_{3v}): the same with $x = 0.185$
2Si(C_{3v}): the same with $x = 0.352$
Hexagonal axes: $a = 3.88$, $c = 30.4$; $A = 18$
Co-ordinates: (000; $\frac{2}{3}\,\frac{1}{3}\,\frac{1}{3}$; $\frac{1}{3}\,\frac{2}{3}\,\frac{2}{3}$) +6Ca($C_{3v}$): $\pm(00x)$; $x = 0.083$
+6Si(C_{3v}): the same with $x = 0.185$
+6Si(C_{3v}): the same with $x = 0.352$

*C*14 ($MgZn_2$ type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $c = 5.223(1)$, $c = 8.566(3)$; $A = 12$
Co-ordinates: 4Mg(C_{3v}): $\pm(\frac{1}{3}\,\frac{2}{3}z;\ \frac{1}{3},\ \frac{2}{3},\ \frac{1}{2} - z)$; $z \approx \frac{1}{16} = 0.062$
2Zn(D_{3d}): 000; $00\frac{1}{2}$
6Zn(C_{2v}): $\pm(x, 2x, \frac{1}{4};\ \bar{2}\bar{x}, \bar{x}, \frac{1}{4};\ x\bar{x}\frac{1}{4})$; $x \approx -\frac{1}{6} = -0.170$

*C*15 ($MgCu_2$ type)

Cubic: O_h^7–$Fd3m$; $a = 7.048$; $A = 24$
Co-ordinates: (000; $\frac{1}{2}\,\frac{1}{2}0$;) + 8Mg($T_d$): 000; $\frac{1}{4}\,\frac{1}{4}\,\frac{1}{4}$
+ 16Cu(D_{3d}): $\frac{5}{8}\,\frac{5}{8}\,\frac{5}{8}$; $\frac{7}{8}\,\frac{7}{8}\,\frac{5}{8}$;

*C*16 ($CuAl_2$ type)

Tetragonal: D_{4h}^{18}–$I4/mcm$; $a = 6.063(3)$, $c = 4.872(3)$; $A = 12$
Co-ordinates: (000; $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}$) +4Cu($D_4$): $\pm(00\frac{1}{4})$
+ 8Al(C_{2v}): $\pm(x, \frac{1}{2} + x, 0;\ \frac{1}{2} + x, \bar{x}, 0)$; $x = 0.158$

	$AuNa_2$	$MnSn_2$	$FeSn_2$	$CoSn_2$	$RhSn_2$	$TiSb_2$	VSb_2	Ta_2B	Mo_2B	W_2B	Mn_2B
x	0.160	0.159	0.160	0.116	0.161	0.158	0.158	0.167	0.170	0.170	0.163

For the compounds $FeGe_2$, (Rh,Pd,Au)Pb_2 no deviation from $x = 0.158$ has been reported.

*C*18 [Marcasite (FeS_2) type]

Orthorhombic: D_{2h}^{12}–$Pnnm$; $a = 4.441$, $b = 5.425$, $c = 3.387$; $A = 6$
Co-ordinates: 2Fe(C_{2h}): 000; $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}$
4S(C_s): $\pm(xy0)$; $\pm(\frac{1}{2} + x, \frac{1}{2} - y, \frac{1}{2})$; $x = 0.20$; $y = 0.38$
For $FeAs_2$: $x = 0.18$; $y = 0.36$
For $NiAs_2$: $x = 0.22$; $y = 0.37$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

For $SeSb_2$: $x = 0.18$; $y = 0.36$
For FeP_2: $x = 0.16$; $y = 0.37$
For $CoTe_2$: $x = 0.22$; $y = 0.36$
For $FeTe_2$: $x = 0.22$; $y = 0.36$
For $FeSe_2$: $x = 0.21$; $y = 0.37$

C22 (Fe_2P type)

Hexagonal: $P\bar{6}2m$; $a = 5.8644, c = 3.4560; A = 9$
Co-ordinates: $3Fe(C_2)$: $\bar{x}00; 0\bar{x}0; xx0; x = 0.26$
$3Fe(C_2)$: $x0\frac{1}{2}; 0x\frac{1}{2}; \bar{x}\bar{x}\frac{1}{2}; x = 0.40$
$1P(D_3)$: $00\frac{1}{2}$
$2P(C_3)$: $\pm(\frac{1}{3}\frac{2}{3}z); z \approx \frac{1}{8} = 0.125$

C23 ($PbCl_2$ type)

Orthorhombic: D_{2h}^{16}–$Pmnb$; $a = 4.534, b = 7.622, c = 9.044; A = 12$
Co-ordinates: $4Pb(C_s)$: $\pm(\frac{1}{4}yz); \pm(\frac{1}{4})\frac{1}{2} + y, \frac{1}{2} - z); y = 0.246; z = 0.905$
$4Cl(C_s)$: the same with $y = 0.85; z = 0.93$
$4Cl(C_s)$: the same with $y = 0.95; z = 0.33$

For Co_2P, Ni_2Si* and ThS_2 the parameters are:

	P	Co	Co	Th	S	S	Si	Ni	Ni
y	0.250	0.862	0.970	0.250	0.850	0.965	0.236	0.825	0.958
z	0.900	0.930	0333	0.875	0.942	0.320	0.886	0.937	0.297

C32 (AlB_2 type)

Hexagonal: D_{6h}^{1}–$P6/mmm$; $a = 3.005, c = 3.257; A = 3$
Co-ordinates: $1Al(D_{6h})$: 000
$2B(D_{3h})$: $\frac{1}{3}\frac{2}{3}\frac{1}{3}; \frac{2}{3}\frac{1}{3}\frac{1}{2}$

C33 (Bi_2Te_2S type)

Rhombohedral: D_{3d}^{5}–$R\bar{3}m$; $a = 10.31; \alpha = 24°10'; A = 5$
Co-ordinates: $2Bi(C_{3v})$: $\pm(xxx); x = 0.392$
$2Te(C_{3v})$: the same with $x = 0.788$
$1S(D_{3d})$: 000

C34 [Calaverite ($AuTe_2$) type]

Monoclinic: C_{2h}^{3}–$C2/m$; $a = 7.1947(4), b = 4.4146(2), c = 5.0703(3); \beta = 90.038(4); A = 6$
Co-ordinates: $(000; \frac{1}{2}\frac{1}{2}0) + 2Au(C_{2h})$: 000
$+4Te(C_s)$: $\pm(x0z); x = 0.689; z = 0.280$

C36 ($MgNi_2$ type)

Hexagonal: D_{6h}^{4}–$P6_3/mmc$; $a = 4.807, c = 15.77; A = 24$
Co-ordinates: $4Mg(C3v)$: $+(\frac{12}{33}z); \pm(\frac{2}{3}, \frac{1}{3}, \frac{1}{2} + z); z \approx \frac{27}{32}$
$4Mg(C_{3v})$: $\pm(00z); \pm(0, 0, \frac{1}{2} + z); z \approx \frac{3}{32}$
$6Ni(C_{2h})$: $\frac{1}{2}00; 0\frac{1}{2}0; \frac{1}{2}\frac{1}{2}0; \frac{1}{2}0\frac{1}{2}; 0\frac{1}{2}\frac{1}{2}; \frac{1}{2}\frac{1}{2}\frac{1}{2}$
$6Ni(C_{2v})$: $\pm(x, 2x, \frac{1}{4}; 2\bar{x}, \bar{x}, \frac{1}{4}; x, \bar{x}, \frac{1}{4}); x \approx \frac{1}{6}$
$4Ni(C_{3v})$: $\pm(\frac{1}{3}\frac{2}{3}z); \pm(\frac{2}{3}, \frac{1}{3}, \frac{1}{2} + z); z \approx \frac{1}{8}$

C37 (Co_2Si type)

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 7.104, b = 4.919, c = 3.725; A = 12$
Co-ordinates: $4Si(C_s)$: $\pm(xy\frac{1}{4}); \pm(\frac{1}{2} - x, \frac{1}{2} + y, \frac{1}{4}); x = 0.440; y = 0.070$
$4Co(C_s)$: the same with $x = 0.103; y = 0.090$
$4Co(C_s)$: the same with $x = 0.772; y = 0.193$
With a different choice of axes and origin, a similarity to the C23 ($PbCl_2$ type) becomes apparent:
Orthorhombic: D_{2h}^{16}–$Pmnb$; $a = 3.73, b = 4.91, c = 7.10; A = 12$

* If Toman's[199] description of δ-Ni_2Si is charged from *Pbnm* to *Pmnb*, and if the origin is chosen differently, this compound belongs to the C23 type.

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

Co-ordinates: 4Si(C_s): $\pm(\frac{1}{4}yz); \pm(\frac{1}{4}, \frac{1}{2}+y, \frac{1}{2}-z); y=0.07; z=0.94$
4Co(C_s): the same with $y=0.59; z=0.897$
4Co(C_s): the same with $y=0.693; z=0.228$

With this orientation the lattice constants are virtually identical with those of δ-Ni_2Si ($a=3.73, b=4.99, c=7.03$) which has the *C*23 type. A redetermination of the Co_2Si type might lead to a still closer similarity than that appearing in this description.

C38 (Cu_2Sb type)

Tetragonal: D^7_{4h}–$P4/nmm$; $a=4.0014, c=6.1044; A=6$
Co-ordinates: 2Cu(D_{2d}): $000, \frac{1}{2}\frac{1}{2}0$
2Cu(C_{4v}): $0\frac{1}{2}z; \frac{1}{2}0\bar{z}; z=0.27$
2Sb(C_{4v}): the same with $z=0.70$

For $Cu_{2-x}Te$; $z(Cu)=0.27$; $z(Te)=0.715$, with vacant sites in the $(0\frac{1}{2}z)$-position.
For Fe_2As and Cr_2As: $z(As)=0.735$; $z(Fe\text{ or }Cr)=0.33$. For CuMgAs: $z(As)=0.75$; $z(Cu, Mg)=0.335$.
For $AlNaSi_4$: $z(Si)=0.79$; $z(Na, Al)=0.37$.

A very similar arrangement is shown by $CuGa_2$ with $a=2.83, c=5.84$. The 3 atoms per cell take the position 000; $\frac{1}{2}\frac{1}{2}x; \frac{1}{2}\frac{1}{2}y$ with $x=0.70; y=0.27$. For comparison a larger cell with $a'=a\sqrt{2}$ can be chosen:

Tetragonal: C^1_{4v}–$P4\,mm$; $a'=4.03, c=5.84; A=6$
Co-ordinates: 2Ga(C_{4v}): $000; \frac{1}{2}\frac{1}{2}0$
2Ga(C_{4v}): $0\frac{1}{2}z; \frac{1}{2}0z; z=0.70$
2Cu(C_{4v}): $0\frac{1}{2}z; \frac{1}{2}0z; z=0.27$

C40 ($CrSi_2$ type)

Hexagonal: D^4_6(or D^5_6)–$P6_122$ (or $P6_422$); $a=4.430(2), 6.365(5); A=9$
Co-ordinates: 3Cr(D_2): $\frac{1}{2}00; \frac{1}{2}\frac{1}{2}\frac{1}{3}; 0\frac{1}{2}\frac{2}{3}$
6Si(C_2): $\pm(x, 2x, 0); x\bar{x}\frac{1}{3}; \bar{x}x\frac{1}{3}; 2x, x, \frac{2}{3}; 2\bar{x}, \bar{x}\frac{2}{3}; x\approx\frac{1}{6}$

C42 (SiS_2 type)

Orthorhombic: D^{26}_{2h}–$Icma$; $a=5.564(2); b=5.552(2); c=9.545(3); A=12$
Co-ordinates: $(000, \frac{1}{2}\frac{1}{2}\frac{1}{2})+4Si(D_2)$: $\pm(0\frac{1}{4}0)$
$+8S(C_s)$: $\pm(x0z); \pm(x\frac{1}{2}\bar{z}); x=0.208; z=0.119$

C44 (GeS_2 type)

Orthorhombic: C^{19}_{2v}–$Fdd2$; $a=11.691(8), b=22.341(3); c=6.68(1); A=72$
Co-ordinates: $(000; \frac{1}{2}\frac{1}{2}0)+8Ge(C_2)$: $000; \frac{1}{4}\frac{1}{4}\frac{1}{4}$
$+16Ge(C_1)$: $xyz; x\bar{y}z; (\frac{1}{4}-x)(\frac{1}{4}+y)(\frac{1}{4}+z); (\frac{1}{4}+x)(\frac{1}{4}-y)(\frac{1}{4}+z)$ with $x=0.125; y=0.139; z=0.00$
$+16S(C_1)$: the same with $x=0.022; y=0.081; z=0.183$
$+16S(C_1)$: the same with $x=0.153; y=-0.014; z=-0.183$
$+16S(C_1)$: the same with $x=0.063; y=0.125; z=-0.278$

C46 [Krennerite ($AuTe_2$) type]

Orthorhombic: C^4_{2v}–$Pma2$; $a=16.51(3), b=8.80(3), c=4.45(3); A=24$
Co-ordinates: 2Au(C_2): $00z; \frac{1}{2}0z; z=0$
2Au(C_s): $\frac{1}{4}yz; \frac{3}{4}\bar{z}; y=0.319; z=0.014$
4Au(C_1): $xyz; \bar{x}\bar{y}z; (\frac{1}{2}-x)\bar{y}z; (\frac{1}{2}+x)\bar{y}z; x=0.124; y=0.666; z=0.500$
2Te(C_s): $\frac{1}{4}yz; \frac{3}{4}\bar{y}z; y=0.018; z=0.042$
2Te(C_s): the same with $y=0.617; z=0.042$
4Te(C_1): as 4Au (C_1) with $x=0.003; y=0.699; z=0.042$
4Te(C_1): the same with $x=0.132; y=0.364; z=0.500$
4Te(C_1): the same with $x=0.119; y=0.964; z=0.500$

C49 ($ZrSi_2$ type)

Orthorhombic: D^{17}_{2h}–$Cmcm$; $a=3.725, b=14.774, c=3.664; A=12$
Co-ordinates: $(000; \frac{1}{2}\frac{1}{2}0)$ $+4Zr(C_{2v})$: $\pm(0y\frac{1}{4}); y=0.106$
$+4Si(C_{2v})$: the same with $y=0.750$
$+4Si(C_{2v})$: the same with $y=0.355$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$C54$ ($TiSi_2$ type)

Orthorhombic: D_{2h}^{24}–$Fddd$; $a = 8.2671(9), b = 4.800(5), c = 8.5505(11); A = 24$
Co-ordinates: (000, $\frac{1}{2}\frac{1}{2}0$; ♪) +8Ti(D_2): 000; $\frac{1}{4}\frac{1}{4}\frac{1}{4}$
+16Si(C_2): $\pm(x00); (\frac{1}{4}+x)\frac{1}{4}\frac{1}{4}; (\frac{1}{4}-x)\frac{1}{4}\frac{1}{4}; x \approx \frac{1}{3}$

C_a(Mg_2Ni type)

Hexagonal: $D_6^4(D_6^5$–$P6_222(P6_422); a = 5.205(1), c = 13.236(2); A = 18$
Co-ordinates: 3Ni(D_2): $00\frac{1}{6}; 00\frac{3}{6}; 00\frac{5}{6}$
3Ni(D_2): $0\frac{1}{2}\frac{1}{6}; \frac{1}{2}0\frac{3}{6}; \frac{1}{2}\frac{1}{2}\frac{5}{6}$
6Mg(C_2): $\pm(\frac{1}{2}0z), \frac{1}{2}\frac{1}{2}(\frac{1}{3}+z); \frac{1}{2}\frac{1}{2}(\frac{1}{3}-z); 0\frac{1}{2}(\frac{2}{3}+z); 0\frac{1}{2}(\frac{2}{3}-z); z = \frac{1}{9}$
6Mg(C_2): $\pm(x, 2x, 0); x\bar{x}\frac{1}{3}; \bar{x}x\frac{1}{3}; 2x, x, \frac{2}{3}; 2\bar{x}, \bar{x}\frac{2}{3}; x = \frac{1}{6}$

C_b(Mg_2Cu type)

Orthorhombic: D_{2h}^{24}–$Fddd$; $a = 5.284, b = 9.07, c = 18.25; A = 48$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♪) +16Cu(C_2): $\pm(00z); \frac{1}{4}\frac{1}{4}(\frac{1}{4}+z); \frac{1}{4}\frac{1}{4}(\frac{1}{4}-z); z = 0.128$
+16 Mg(C_2): the same with $z = 0.411$
+16 Mg(C_2): $\pm(0y0; \frac{1}{4}(\frac{1}{4}+y)\frac{1}{4}; \frac{1}{4}(\frac{1}{4}-y)\frac{1}{4}; y = 0.161$

C_c($ThSi_2$ type)

Tetragonal: D_{4h}^{19}–$I4_1/amd$; $a = 4.107, c = 14.07; A = 12$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +4Th($D_{2d}$): 000; $0\frac{1}{2}\frac{1}{4}$
+8Si(C_{2v}): $\pm(00z); 0\frac{1}{2}(\frac{1}{4}+z); 0\frac{1}{2}(\frac{1}{4}-z); z = 0.417$

C_e($CoGe_2$ type)

Orthorhombic: C_{2v}^{17}–Aba; $a \approx b = 5.670, c = 10.796(3); A = 24$
Co-ordinates: (000, $0\frac{1}{2}\frac{1}{2}$) +4Co(C_2): $00z; \frac{1}{2}\frac{1}{2}z; z = -0.012$
+4Co(C_2): the same with $z = -0.238$
+8Ge(C_1): $xyz; \bar{x}\bar{y}z; (\frac{1}{2}-x)(\frac{1}{2}+y)z; (\frac{1}{2}+x)(\frac{1}{2}-y)z$; with
$x = 0.342; y = 0.158; z = -\frac{1}{8}$
+8Ge(C_1): the same with $x = y = \frac{1}{4}; z = \frac{1}{8}$

The real composition is $Co_{0.9}\ Ge_2$.

C_f

Tetragonal: a series of types composed of alternating sheets of the Fluorite—and the $C16(Al_2Cu)$—type. Whereas the length of the a-axis is approximately constant, the length of the c-axis varies due to the different possibilities of the sheet sequences.

	$RhSn_2$	$PdSn_2$
a	6.38	6.55
c	17.88	24.57

C_g(ThC_2 type)

Monoclinic: C_{2h}^6–$C2/c$; $a = 6.684(2), b = 4.220(1), c = 6.73(5), \beta = 103.91(1)°; A = 12$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$) +4Th(C_2): $\pm(0y\frac{1}{4}); y = 0.202$
+8C(C_1): $\pm(xyz); \pm(x, \bar{y}, \frac{1}{2}+z)$; with $x = 0.29; y = 0.13; z = 0.08$

C_h (Cu_2Te type)

Hexagonal: D_{6h}^1–$P6/mmm$; $a = 4.188, c = 7.251; A = 6$
Co-ordinates: 2Te(C_{6v}): $\pm(00z); z = 0.306$
4Cu(C_{3v}): $\pm(\frac{1}{3}\frac{2}{3}z); \pm(\frac{1}{3}\frac{2}{3}\bar{z}); z = 0.160$

$D0_2$ ($CoAs_3$ type)

Cubic: T_h^5–$Im3$; $a = 8.208; A = 32$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +8Co($C_{3i}$): $\frac{1}{4}\frac{1}{4}\frac{1}{4}; \frac{3}{4}\frac{3}{4}\frac{1}{4}$; ♪
+24As(C_s): $\pm(xy0;$ ♪$) \pm(x\bar{y}0;$ ♪$); x = 0.35; y = 0.15$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D0_3$ (BiF_3 or $BiLi_3$ type)

Cubic: 0_h^5–$Fm3m$; for $BiLi_3$: $a = 6.708$; $A = 16$ (for BiF_3 $a = 5.853(6)$)
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♪) +4Bi(O_h): 000
+4Li(O_h): $\frac{1}{2}\frac{1}{2}\frac{1}{2}$
+8Li(T_d): $\pm(\frac{1}{4}\frac{1}{4}\frac{1}{4})$

$D0_9$(ReO_3 or Cu_3N type)

Cubic: O_h^1–$Pm3m$; for Cu_3N; $a = 3.813$; $A = 4$ (for ReO_3 $a = 3.74774(4)$)
Co-ordinates: IN(O_h): 000; 3Cu(D_{4h}): $\frac{1}{2}00$; ♪

$D0_{11}$ (Fe_3C type)

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 4.523$, $b = 5.090$, $c = 6.748$; $A = 16$
Co-ordinates: 4Fe(C_s): $\pm(xy\frac{1}{4})$; $\pm(\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{4})$; $x = 0.833$; $y = 0.040$
8Fe(C_1): $\pm(xyz)$; $x, y, \frac{1}{2}-z$; $\pm(\frac{1}{2}-x, \frac{1}{2}+y, z$; $\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{2}-z)$;
$x = 0.333$; $y = 0.183$; $z = 0.065$
4C(C_s): $\pm(xy\frac{1}{4})$; $\pm(\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{4})$; $x = 0.47$; $y = 0.86$

$D0_{18}$ (Na_3As type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 5.088$, $c = 8.982$; $A = 8$
Co-ordinates: 2As(D_{3h}): $\pm(\frac{1}{3}\frac{2}{3}\frac{1}{4})$
2Na(D_{3h}): $\pm(00\frac{1}{4})$
4Na(C_{3v}): $\pm(\frac{1}{3}\frac{2}{3}z; \frac{2}{3}, \frac{1}{3}, \frac{1}{2}+z)$; $z = 0.583$

$D0_{19}$ (Mg_3Cd type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 6.310$, $c = 5.080$; $A = 8$
Co-ordinates: 2Cd(D_{3h}): $\pm(\frac{1}{3}\frac{2}{3}\frac{1}{4})$
6Mg(C_{2v}): $\pm(2x, x, \frac{1}{4}; \bar{x}x\frac{1}{4}; \bar{x}, 2x, \frac{1}{4}; x \approx \frac{1}{6}$

$D0_{20}$ ($NiAl_3$ type)

Orthorhombic: D_{2h}^{16}–$Pnma$; $a = 6.618(1)$, $b = 7.368(1)$, $c = 4.814(1)$; $A = 16$
Co-ordinates: 4Ni(C_s): $\pm(x\frac{1}{4}z)$; $\pm(\frac{1}{2}+x, \frac{1}{4}, \frac{1}{2}-z)$; $x = -0.731$; $z = -0.055$
4Al(C_s): the same with $x = 0.011$; $z = 0.415$
8Al(C_1): $\pm(xyz)$; $\pm(\frac{1}{2}+x, \frac{1}{2}-y, \frac{1}{2}-z)$; $\pm(x, \frac{1}{2}-y, z)$; $\pm(\frac{1}{2}+x, y, \frac{1}{2}-z)$
with $x = 0.174$; $y = 0.053$; $z = 0.856$

Following the choice of co-ordinates and of the origin in the original paper, $NiAl_3$ was described in the Strukturbericht as a separate type. However, another choice of co-ordinates and of origin shows that the structure is very similar to the $D0_{11}$ (Fe_3C) type:

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 4.814(1)$, $b = 6.618(1)$, $c = 7.368(1)$, $A = 16$
Co-ordinates: 4Al(C_s): $\pm(xy\frac{1}{4})$; $\pm(\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{4})$; $x = 0.915$; $y = 0.011$
8Al(C_1): $\pm(xyz, xy\frac{1}{2}-z)$; $\pm(\frac{1}{2}-x, \frac{1}{2}+y, z$; $\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{2}-z)$; $x = 0.356$;
$y = 0.174$; $z = 0.053$
4Ni(C_s): $\pm(xy\frac{1}{4})$; $\pm(\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{4})$; $x = 0.445$; $y = 0.869$

$D0_{21}$(Cu_3P type)

Hexagonal: D_{3d}^4–$P6_3\,cm$; $a = 6.88$, $c = 7.18$; $A = 24$
Co-ordinates: 6P(C_2): $\pm(x0\frac{1}{4})$; $\pm(0x\frac{1}{4})$; $\pm(\bar{x}\bar{x}\frac{1}{4})$; $x = 0.38$
2Cu(C_{3i}): $000, 00\frac{1}{2}$
4Cu(C_3): $\pm(\frac{1}{3}\frac{2}{3}z)$; $\pm(\frac{1}{3}, \frac{2}{3}, \frac{1}{2}+z)$; $z = 0.17$
12Cu(C_1): $\pm[xyz; \bar{y}(x-y)z; (y-x)\bar{x}z; \bar{y}\bar{z}(\frac{1}{2}+z); x(x-y)(\frac{1}{2}+z); (x-y)y(\frac{1}{2}+x)]$;
$x = 0.69$; $y = 0.07$; $z = 0.08$

See remarks by Haraldsen[200]

$D0_{22}$($TiAl_3$ type)

Tetragonal: D_{4h}^{17}–$I4/mmm$; $a = 3.846$, $c = 8.594$; $A = 8$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +2Ti($D_{4h}$): 000
+2Al(D_{4h}): $00\frac{1}{2}$
+Al(D_{2d}): $0\frac{1}{2}\frac{1}{4}$; $\frac{1}{2}0\frac{1}{4}$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D0_{23}$($ZrAl_3$ type)

Tetragonal: D_{4h}^{17}–$I4/mmm$; $a = 4.0074(6), c = 17.286(4); A = 16$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +4Zr($C_{4v}$): $\pm(00z); z = 0.122 \approx \frac{1}{8}$
+4Al(D_{2h}): $0\frac{1}{2}0; \frac{1}{2}00$
+4Al(D_{2d}): $0\frac{1}{2}\frac{1}{4}; \frac{1}{2}0\frac{1}{4}$
+4Al(C_{4v}): $\pm(00z); z = 0.361 \approx \frac{3}{8}$

$D0_{24}$($TiNi_3$ type)

Hexagonal: D_{6h}^{4}–$P6_3/mmc$; $a = 5.102\,8, c = 8.271\,9; A = 16$
Co-ordinates: 2Ti(D_{3d}): $000; 00\frac{1}{2}$; 2Ti(D_{3h}): $\frac{1}{3}\frac{2}{3}\frac{1}{4}; \frac{2}{3}\frac{1}{3}\frac{3}{4}$
6Ni(C_{2h}): $\frac{1}{2}00; 0\frac{1}{2}0; \frac{1}{2}\frac{1}{2}0; \frac{1}{2}0\frac{1}{2}; 0\frac{1}{2}\frac{1}{2}; \frac{1}{2}\frac{1}{2}\frac{1}{2}$
6Ni(C_{2v}): $\pm(x, 2x\frac{1}{4}; 2\bar{x}, \bar{x}, \frac{1}{4}; x\bar{x}\frac{1}{4}); x = \approx -\frac{1}{6}$

$D0_a$(β-$TiCu_3$ type)

Orthorhombic: D_{2h}^{13}–$Pmmn$; $a = 5.169, b = 4.342, c = 4.531; A = 8$
Co-ordinates: 2Ti(C_{2v}): $00z; \frac{1}{2}\frac{1}{2}\bar{z}; z = 0.655$
2Cu(C_{2v}): $0\frac{1}{2}z; \frac{1}{2}0\bar{z}; z = 0.345$
4Cu(C_s): $x0z; x0z; (\frac{1}{2}+x)\frac{1}{2}\bar{z}; (\frac{1}{2}-x)\frac{1}{2}\bar{z}; x = \frac{1}{4}; z = 0.155$

$D0_c$ (U_3Si type)

Tetragonal: D_{4h}^{18}–$I4/mcm$; $a = 6.032\,8(4), 8.690\,7(6); A = 16$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$ +4U(D_4): $\pm(00\frac{1}{4})$
+8U(C_{2v}): $\pm(x, \frac{1}{2}+x, 0); \pm(\frac{1}{2}+x, \bar{x}, 0); x = 0.231$
+4Si(D_{2d}): $0\frac{1}{2}\frac{1}{4}; \frac{1}{2}0\frac{1}{4}$

$D0_d$ (Mn_3As type)

Orthorhombic: D_{2h}^{13}–$Pmmn$; $a \approx b = 3.788, c = 16.29; A = 16$
Co-ordinates: 3 times 2Mn(C_{2v}): $00z; \frac{1}{2}\frac{1}{2}z$;
2As(C_{2v}): the same
3 times 2Mn(C_{2v}): $0\frac{1}{2}z; \frac{1}{2}0\bar{z}$;
2As (C_{2v}): the same

	(00*z*)-*position*				(0$\frac{1}{2}$*z*)-*position*			
	Mn *I*	Mn *II*	Mn *III*	As *I*	Mn *IV*	Mn *V*	Mn *VI*	As *II*
z	0.194	−0.194	−0.435	0.409	0.307	−0.307	−0.066	0.091

$D1_3$ ($BaAl_4$ type)

Tetragonal: D_{4h}^{17}–$I4/mmm$; $a = 4.566, c = 11.250; A = 10$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +2Ba($D_{4h}$): 000
+4Al(C_{4v}): $\pm(00z); z = 0.380$
+Al(D_{2d}): $0\frac{1}{2}\frac{1}{4}; \frac{1}{2}0\frac{1}{4}$

$D1_a$ ($MoNi_4$ type)

Tetragonal: C_{4h}^{5}–$I4/m$; $a = 5.683, c = 3.592; A = 10$
Co-ordinates: (000, $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +2Mo($C_{4h}$): 000
+8Ni(C_s): $\pm(xy0); \pm(y\bar{x}0); x = 0.400; y = 0.200$

$D1_b$ (UAl_4 type)

Orthorhombic: C_{2v}^{20} (or D_{2h}^{28}–$I2ma$ (or $Imma$); $a = 4.42, b = 6.30, c = 13.63; A = 20$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +4U($C_{2v}$): $\pm(0\frac{1}{4}z); z = 0.111$
+4Al(C_{2v}): the same with $z = -0.111$
+4Al(C_{2v}): $000, 0\frac{1}{2}\frac{1}{2}$
+8Al(C_s): $\pm(0yz); \pm(0, \frac{1}{2}-y, z); y = -0.033; z = 0.314$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D1_c$ ($PtSn_4$ type)

Orthorhombic: C_{2v}^{17}–$Aba2$; $a = 6.382\,3(1), b = 6.419\,0(1), c = 11.366\,6(2); A = 20$
Co-ordinates: (000; $0\frac{1}{2}\frac{1}{2}$) +4Pt(C_2): $00z$; $\frac{1}{2}\frac{1}{2}z; z = 0$
+2 times 8Sn(C_1): $xyz; \bar{x}\bar{y}z; (\frac{1}{2}+x)(\frac{1}{2}-y)z; (\frac{1}{2}-x)(\frac{1}{2}+y)z$

	x	y	z
Sn I	0.173	0.327	0.125
Sn II	0.33	0.17	−0.13

$D1_d$ ($PtPb_4$ type)

Tetragonal: D_{4h}^{3}–$P4/nbm$; $a = 6.666(10), c = 5.978(10); A = 10$
Co-ordinates: 2Pt(D_4): $000; \frac{1}{2}\frac{1}{2}0$
8Pb(C_s): $x(\frac{1}{2}+x)z; \bar{x}(\frac{1}{2}-x)z; (\frac{1}{2}+x)\bar{x}z; (\frac{1}{2}-x)xz; x(\frac{1}{2}-x)\bar{z}; \bar{x}(\frac{1}{2}+x)\bar{z}; (\frac{1}{2}+x)x\bar{z}$;
$(\frac{1}{2}-x)\bar{x}\bar{z}; x = 0.175; z = 0.255$

$D1_e$ (UB_4 type)

Tetragonal: D_{4h}^{5}–$P4/mbm$; $a = 7.077, c = 3.983; A = 20$
Co-ordinates: 4U(C_{2v}): $\pm(x, \frac{1}{2}+x, 0; \frac{1}{2}+x, \bar{x}, 0); x = 0.31$
4B(C_4): $\pm(00z; \frac{1}{2}\frac{1}{2}z); z = 0.2$
4B(C_{2v}): $\pm(x, \frac{1}{2}+x, \frac{1}{2}; \frac{1}{2}+x, \bar{x}, \frac{1}{2}); x = 0.1$
4B(C_s): $\pm[xy\frac{1}{2}; y\bar{x}\frac{1}{2}; (\frac{1}{2}+x)(\frac{1}{2}-y)\frac{1}{2}; (\frac{1}{2}+y)(\frac{1}{2}+x)\frac{1}{2}]\, x = 0.2; y = 0.04$

$D1_f$ (Mn_4B type)

Orthorhombic: D_{2h}^{24}–$Fddd$; $a = 14.53, b = 7.293, c = 4.209; A = 40$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♩) +16Mn(C_2): $\pm(0y0); \frac{1}{4}(\frac{1}{4}+y)\frac{1}{4}; \frac{1}{4}(\frac{1}{4}-y)\frac{1}{4}; y = 0.333$
+16Mn(C_2): $\pm(x00); (\frac{1}{4}+x)\frac{1}{4}\frac{1}{4}; (\frac{1}{4}-x)\frac{1}{4}\frac{1}{4}; x = 0.083$
+8B(C_2): at random in the same position with $x = 0.375$

$D1_g$ (B_4C type)

Rhombohedral: D_{3d}^{5}–$R\bar{3}m$; $a = 5.19; \alpha = 66°18'; A = 15$
(Hexagonal setting: $a = 5.60, c = 12.12; A = 45$)
Co-ordinates: (000; $\frac{1}{3}\frac{2}{3}\frac{1}{3}$; $\frac{2}{3}\frac{1}{3}\frac{2}{3}$) +3C($D_{3d}$): $00\frac{1}{2}$ + 6C(C_{3v}): $\pm(00z); z = 0.385$
+18B(C_1): $\pm(x\bar{x}z); \pm(x, 2x, z); \pm(2\bar{x}, \bar{x}, z); x = 0.106; z = 0.113$
+18B(C_1): the same with $x = \frac{1}{6}; z = 0.360$

$D2_1$ (CaB_6 type)

Cubic: O_h^1–$Pm3m$; $a = 4.151\,2; A = 7$
Co-ordinates: 1Ca(O_h): 000
6B(C_{4v}): $\pm(\frac{1}{2}\frac{1}{2}x;$ ♩$); x = 0.20$

$D2_3$ ($NaZn_{13}$ type)

Cubic: O_h^6–$FM\bar{3}c$; $a = 12.283\,6(3); A = 112$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♩) +8Na(O): $\pm(\frac{1}{4}\frac{1}{4}\frac{1}{4})$
+8Zn(T_h): $000; \frac{1}{2}\frac{1}{2}\frac{1}{2}$
+96Zn(C_s): $\pm(0yz;$ ♩$; \frac{1}{2}zy;$ ♩$; 0y\bar{z};$ ♩$; \frac{1}{2}\bar{z}y;$ ♩$); y = 0.180\,6$;
$z = 0.119\,2$

For $Be_{13}U$, neutron diffraction gave: $y = 0.178; z = 0.112$

$D2_b$ ($ThMn_{12}$ type)

Tetragonal: D_{4h}^{17}–$I4/mmm$; $a = 8.74(1), 4.95(1); A = 26$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +2Th($D_{4h}$): 000
+8Mn(C_{2h}): $\frac{1}{4}\frac{1}{4}\frac{1}{4}, \frac{3}{4}\frac{3}{4}\frac{1}{4}$; ♩
+8Mn(C_{2v}): $\pm(x00); \pm(0x0); x = 0.361$
+8Mn(C_{2v}): $\pm(x\frac{1}{2}0); \pm(\frac{1}{2}x0); x = 0.277$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D2_c$ (U_6Mn type)

Tetragonal: D_{4h}^{18}–$I4/mcm$ (or subgroup); $a = 10.312(5), b = 5.255(4); A = 28$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +4Mn($D_4$): $\pm(00\frac{1}{4})$
+8U(C_{2v}): $\pm(x, \frac{1}{2}+x, 0); \pm(\frac{1}{2}+x, \bar{x}, 0); x = 0.405$
+16U(C_s): $\pm(xy0; y\bar{x}0; x\bar{y}\frac{1}{2}; yx\frac{1}{2}); x = 0.213; y = 0.103$

$D2_d$ ($CaCu_5$ type)

Hexagonal: D_{6h}^{1}–$P6/mmm$; $a = 5.092, c = 4.086; A = 6$
Co-ordinates: 1Ca(D_{6h}): 000
2Cu(D_{3h}): $\frac{1}{3}\frac{2}{3}0; \frac{2}{3}\frac{1}{3}0$; 3Cu($D_{2h}$): $\frac{1}{2}0\frac{1}{2}; 0\frac{1}{2}\frac{1}{2}; \frac{1}{2}\frac{1}{2}\frac{1}{2}$

$D2_e$ ($BaHg_{11}$ type)

Cubic: O_h^1–$Pm3m$; $a = 9.586; A = 36$
Co-ordinates: 3Ba(D_{4h}): $\frac{1}{2}00$; ♪
1Hg(O_h): 000; 8Hg (C_{3v}): $\pm(xxx; xx\bar{x}$; ♪); $x = 0.155$
12Hg(C_{2v}): $\pm(xx0$; ♪); $\pm(x\bar{x}0$; ♪); $x = 0.345$
12Hg(C_{2v}): $\pm(xx\frac{1}{2}$; ♪); $\pm(x\bar{x}\frac{1}{2}$; ♪); $x = 0.275$

$D2_f$ (UB_{12} type)

Cubic: O_h^5–$Fm3m$; $a = 7.475; A = 52$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$;) ♪) +4U(O_h): 000
+48B(C_{2v}): $\pm(\frac{1}{2}xx$; ♪); $\pm(\frac{1}{2}x\bar{x}$; ♪); $x = \frac{1}{6}$

$D2_g$ (Fe_8N type)

Tetragonal: D_{4h}^{17}–$I4/mmm$; $a = 5.720, c = 6.292; A = 18$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 2N($D_{4h}$): 000
+ 4Fe(D_{2d}): $\frac{1}{2}0\frac{1}{4}; 0\frac{1}{2}\frac{1}{4}$
+ 4Fe(C_{2v}): $\pm(00z); z = 0.56$
+ 8Fe(C_{2v}): $\pm(xx0); \pm(x\bar{x}0); x = \frac{1}{4}$

$D5_2$ (La_2O_3 type)

Hexagonal: D_{3d}^{3}–$P\bar{3}m1$; $a = 3.937, c = 6.130, c/a = 1.56; A = 5$
Co-ordinates: 2La(C_{3v}): $\frac{1}{3}\frac{2}{3}z; \frac{2}{3}\frac{1}{3}z; z \approx 0.23$
1O(D_{3d}): 000; 2O(C_{3v}): $\frac{1}{3}\frac{2}{3}z; \frac{2}{3}\frac{1}{3}\bar{z}; z \approx 0.63$

Apart from the different c/a-values and small differences in the z-values this type is similar to the $D5_{13}$ (Ni_2Al_3) type.

$D5_3$ (Mn_2O_3 type)

Cubic: T_h^7–$Ia3$; $a = 9.414; A = 80$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +8Mn($C_{3i}$): $\frac{1}{4}\frac{1}{4}\frac{1}{4}; \frac{1}{4}\frac{3}{4}\frac{3}{4}$; ♪;
+24Mn(C_2): $\pm(x0\frac{1}{4}$; ♪); $\pm(x\frac{1}{2}\frac{3}{4}$; ♪); $x = -0.030$
+48O(C_1): $\pm(xyz$; ♪); $\pm(x, \bar{y}, \frac{1}{2}-z$; ♪); $\pm(\frac{1}{2}+x, \bar{y}, z$; ♪);
$\pm(x, \frac{1}{2}+y, \bar{z}$; ♪); $x = 0.39; y = 0.15; z = 0.38$

$D5_8$ (Sb_2S_3 type)

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 11.189(5), b = 11.380(5), c = 3.829(2), A = 20$
Co-ordinates: 4Sb(C_s): $\pm(xy\frac{1}{4}); \pm(\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{4}); x = 0.33; y = 0.03$
4Sb(C_s): the same with $x = -0.04; y = -0.15$
4S(C_s): the same with $x = 0.88; y = 0.05$
4S(C_s): the same with $x = -0.44; y = -0.13$
4S(C_s): the same with $x = 0.19; y = 0.21$

For other substances:

	M *I*	M *II*	S *I*	S *II*	S *III*
U_2S_3, x	0.311	−0.008	0.878	−0.439	0.206
y	−0.014	−0.195	0.053	−0.129	0.230
Th_2S_3, x	0.314	−0.019	0.878	−0.439	0.206
y	−0.022	−0.200	0.053	−0.129	0.230

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D5_9$ (Zn_3P_2 type)

Tetragonal: D_{4h}^{15}–$P4/nmc$; $a = 8.10, c = 11.45; A = 40$
Co-ordinates: 4P(C_{2v}): $\pm(00z); \pm(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}+z); z = 0.25$
4P(C_{2v}): $0\frac{1}{2}z; \frac{1}{2}0\bar{z}; 0\frac{1}{2}(\frac{1}{2}+z); \frac{1}{2}0(\frac{1}{2}-z); z = 0.24$
8P(C_s): $\pm(xx0; \bar{x}x0; \frac{1}{2}+x, \frac{1}{2}+x, \frac{1}{2}; \frac{1}{2}-x, \frac{1}{2}+x, \frac{1}{2}); x = 0.26$
8Zn(C_s): $0xz; 0\bar{x}z; x0\bar{z}; \bar{x}0\bar{z}; \frac{1}{2}(\frac{1}{2}+x)(\frac{1}{2}-z); \frac{1}{2}(\frac{1}{2}-x)(\frac{1}{2}-z); (\frac{1}{2}+x)\frac{1}{2}(\frac{1}{2}+z);$
$(\frac{1}{2}-x)\frac{1}{2}(\frac{1}{2}+z); x = 0.22; z = 0.10$
8Zn(C_s): the same with $x = 0.28; z = 0.39$
8Zn(C_s): the same with $x = 0.26; z = 0.65$

$D5_{10}$(Cr_3C_2 type)

Orthorhombic: D_{2h}^{16}–$Pbnm$; $a = 11.471\,9(7), b = 5.532\,9(5), c = 2.829\,0(2); A = 20$
Co-ordinates: 4C(C_s): $\pm(xy\frac{1}{4}); \pm(\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{4}); x = 0.11; y = -0.10$
4C(C_s): the same with $x = -0.06; y = 0.22$
4Cr(C_s): the same with $x = 4.406; y = 0.03$
4Cr(C_s): the same with $x = -0.230; y = 0.175$
4Cr(C_s): the same with $x = -0.070; y = -0.150$

$D5_{13}$ (Ni_2Al_3 type)

Hexagonal: $D_{3d}^3 - P\bar{3}m1; a = 4.036(1), c = 4.897(1), c/a = 1.21; A = 5$
Co-ordinates: 2Ni(C_{3v}): $\frac{1}{3}\frac{2}{3}z; \frac{2}{3}\frac{1}{3}z; z = 0.149$
1Al(D_{3d}): 000; 2Al(C_{3v}): $\frac{1}{3}\frac{2}{3}z; \frac{2}{3}\frac{1}{3}z; z = 0.648$

Apart from the different c/a-values and small differences in the z-values, this type is similar to the $D5_2$ (La_2O_3) type.

For Ni_2Ga_3: $z(Ni) = 0.138; z(Ga) = 0.625$
For Ni_2In_3: $z(Ni) = 0.135; z(In) = 0.641$

$D5_a$ (U_3Si_2 type)

Tetragonal: D_{4h}^5–$P4/mbm$; $a = 7.329\,9(4), c = 3.900\,4(5); A = 10$
Co-ordinates: 2U(C_{4h}): 000; $\frac{1}{2}\frac{1}{2}0$
4U(C_{2v}): $\pm(x, \frac{1}{2}+x, \frac{1}{2}); \pm(\frac{1}{2}+x, \bar{x}, \frac{1}{2}); x = 0.181$
4Si(C_{2v}): $\pm(x, \frac{1}{2}+x, 0); \pm(\frac{1}{2}+x, \bar{x}, \frac{1}{2}); x = 0.389$

$D5_b$ (Pt_2Sn_3 type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 4.334\,5, c = 12.960\,6; A = 10$
Co-ordinates: 4Pt(C_{3v}): $\pm(\frac{1}{3}\frac{2}{3}z); \pm(\frac{1}{2}, \frac{2}{3}, \frac{1}{2}-z); z = 0.14$
4Sn(C_{3v}): the same with $z = -0.07$
2Sn(D_{3h}): $\pm(00\frac{1}{4})$

$D5_c$ (Pu_2C_3 type)

Cubic: T_d^6–$I\bar{4}3d$; $a = 8.132; A = 40$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 16Pu($C_3$): $xxx; (x+\frac{1}{4})(x+\frac{1}{4})(x+\frac{1}{4}); (\frac{1}{2}+x, \frac{1}{2}-x, \bar{x})$; ♪;
$(\frac{3}{4}+x)(\frac{1}{4}-x)(\frac{3}{4}-x)$; ♪; $x = 0.050$
+ 24C(C_2): $y0\frac{1}{4}$; ♪; $(\frac{1}{2}-y)0\frac{3}{4}$; ♪; $(\frac{3}{4}+y)0\frac{3}{4}$; ♪; $(\frac{3}{4}-y)0\frac{1}{4}$; ♪; $y = 0.28$

$D5_e$ (Ni_3S_2 type)

Rhombohedral: D_3^7–$R32; a = 4.08; \alpha = 89°25'; A = 5$
Co-oridinates: 3Ni(C_2): $\frac{1}{2}x\bar{x}$; ♪; $x = \frac{1}{4}$
2S(C_3): $\pm(xxx); x = \frac{1}{4}$

$D5_f$ (As_2S_3 type)

Monoclinic: C_{5h}^2–$P2_1/n; a = 12.191(5), b = 9.577(4), c = 4.256(2); \beta = 109.76(8)°; A = 20$
Co-ordiantes: 4As(C_1): $\pm(xyz); \pm(\frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{2}-z)$
All the other atoms in the same position.

	As *I*	As *II*	S *I*	S *II*	S *III*
x	0.268	0.482	0.410	0.340	0.125
y	0.187	0.313	0.120	0.380	0.305
z	0.161	−0.339	0.454	−0.046	0.455

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D7_1$ (Al_4C_3 type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$; $a = 8.53$; $\alpha = 22°28'$; $A = 7$
(Hexagonal setting: $a = 3.32$, $c = 24.95$; $A = 21$)
Co-ordinates: 1C(D_{3d}): 000
2C(C_{3v}): $\pm(xxx)$; $x = 0.217$
2Al(C_{3v}): the same with $x = 0.293$
2Al(C_{3v}): the same with $x = 0.128$

$D7_2$ (Co_3S_4 type)

Cubic: O_h^7–$Fd\bar{3}m$; $a = 9.404(1)$; $A = 56$
Co-ordiantes: (000; $\frac{1}{2}\frac{1}{2}0$; ♪) + 8Co(T_d): 000; $\frac{1}{4}\frac{1}{4}\frac{1}{4}$
+16Co(D_{3d}): $\frac{5}{8}\frac{5}{8}\frac{5}{8}$; $\frac{7}{8}\frac{7}{8}\frac{7}{8}$; ♪
+32S(C_{3v}): $xxx; x\bar{x}\bar{x}$; ♪; $(\frac{1}{4}-x)(\frac{1}{4}-x)(\frac{1}{4}-x)$; $(\frac{1}{4}-x)(\frac{1}{4}-x)(\frac{1}{4}+x)$; ♪; $x = -0.135$

An ordered variety of this type is described as $H1$ (Spinel) type.

$D7_3$ (Th_3P_4 type)

Cubic: T_d^6–$I\bar{4}3d$; $a = 8.618$; $A = 28$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 12Th($S_4$): $\frac{1}{4}\frac{3}{8}0$; ♪; $\frac{3}{4}\frac{1}{8}0$; ♪
+16P(C_3): xxx; $(\frac{1}{4}+x)(\frac{1}{4}+x)(\frac{1}{4}+x)$; $(\frac{1}{2}+x)(\frac{1}{2}+x)\bar{x}$; ♪; $(\frac{3}{4}+x)(\frac{1}{4}-x)$
$(\frac{3}{4}-x)$; ♪; $x = \approx \frac{1}{12} = 0.083$

In the compounds (La, Ce, Ac, Pu, Am)$_2S_3$, $10\frac{2}{3}$ metal ions occupy the Th positions at random.

$D7_a$ (Ni_3Sn_4 type)

Monoclinic: C_{2h}^3–$C2/m$; $a = 12.214(6)$, $b = 4.060(2)$, $c = 5.219(3)$; $\beta = 105.0(1)°$; $A = 14$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$) + 2Ni(C_{2h}): 000
+4Ni(C_s): $\pm(x0z)$; $x = 0.220$; $z = 0.350$
+4Sn(C_s): the same with $x = 0.428$; $z = 0.675$
+4Sn(C_s): the same with $x = 0.180$, $z = 0.800$

$D7_b$ (Ta_3B_4 type)

Orthorhombic: D_{2h}^{25}–$Immm$; $a = 3.036$, $b = 12.846$, $c = 2.958$; $A = 14$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 2Ta($D_{2h}$): $000\frac{1}{2}$
+4Ta(C_{2v}): $\pm(0y0)$; $y = 0.180$
+4B(C_{2v}): the same with $y = 0.375$
+4B(C_{2v}): $\pm(0y\frac{1}{2})$; $y = 0.444$

$D8_1$ (Fe_3Zn_{10} type)

Cubic: O_h^9–$Im3m$; $a = 9.018(3)$; $A = 52$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 12Fe ($C_{4v}$): $\pm(x00; ♪)$; $x \approx \frac{1}{3}$
+16Zn(C_{3v}): $\pm(xxx; x\bar{x}\bar{x}; ♪)$; $x \approx \frac{1}{6}$
+24Zn(C_{2v}): $\pm(xx0; ♪; x\bar{x}0; ♪)$; $x \approx \frac{1}{3}$

$D8_2$(Cu_5Zn_8 type)

Cubic: T_d^3–$I\bar{4}3m$; $a = 8.859(3)$; $A = 52$
Co-ordiantes: (000, $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 12Cu($C_{2v}$): $\pm(x00; ♪)$; $x = 0.355 \approx \frac{1}{3}$
+8Cu(C_{3v}): $xxx; x\bar{x}\bar{x}$; ♪; $x = -0.172 \approx -\frac{1}{6}$
+8Zn(C_{3v}): the same with $x = 0.110 \approx +\frac{1}{6}$
+24Zn(C_s): xxz; ♪; $x\bar{x}\bar{z}$; ♪; $\bar{x}\bar{x}z$; ♪; $\bar{x}x\bar{z}$; ♪; $x = 0.313 \approx \frac{1}{3}$;
$z = 0.036 \approx 0$

$D8_3$ (Cu_9Al_4 type)

Cubic: T_d^1–$P\bar{4}3m$; $a = 8.7023(5)$; $A = 52$
Co-ordinates: 6Cu(C_{2v}): $\pm(x00; ♪)$; $x = 0.356 \approx \frac{1}{3}$
6Cu(C_{2v}): $+(x\frac{1}{2}\frac{1}{2}; ♪)$; $x = 0.856 \approx \frac{1}{2} + \frac{1}{3}$
4Cu(C_{3v}): $xxx; x\bar{x}\bar{x}$; ♪; $x = -0.172 \approx -\frac{1}{6}$
4Cu(C_{3v}): the same with $x = 0.331 \approx \frac{1}{2} - \frac{1}{6}$
4Cu(C_{3v}): the same with $x = 0.601 \approx \frac{1}{2} + \frac{1}{6}$
4Al(C_{3v}): the same with $x = 0.112 \approx \frac{1}{6}$
12Al(C_s): xxz; ♪; $x\bar{x}\bar{z}$; ♪; $\bar{x}\bar{x}z$; ♪; $\bar{x}x\bar{z}$; $x = 0.812 \approx \frac{1}{2} + \frac{1}{3}$; $z = 0.536 \approx \frac{1}{2}$
12Cu(C_s): the same with $x = 0.312 \approx \frac{1}{3}$; $z = 0.036 \approx 0$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D8_4$ ($Cr_{23}C_6$ type)

Cubic: O_h^5–$Fm3m$; $a = 10.650(2)$; $A = 116$
Co-oridiantes: (000; $\frac{1}{2}\frac{1}{2}0$; ♪) + 4Cr(O_h): 000
+8Cr(T_d): $\pm(\frac{1}{4}\frac{1}{4}\frac{1}{4})$
+32Cr(C_{3v}): $\pm(xxx; x\bar{x}\bar{x}; ♪)$; $x = 0.385$
+48Cr(C_{2v}): $\pm(xx0; ♪; x\bar{x}0; ♪)$; $x = 0.165$
+24C(C_{4v}): $\pm(x00; ♪)$; $x = 0.275$

$D8_5$ (Fe_7W_6 type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$; $a = 9.02$; $\alpha = 30°31'$; $A = 13$
(Hexagonal setting: $a = 4.74, c = 25.75$; $A = 39$)
Co-ordiantes: 1Fe(D_{3d}): 000
6Fe(C_s): $\pm(xxz; ♪)$; $x = 0.09$; $z = 0.59$
2W(C_s): $\pm(xxx)$; $x = \frac{1}{6} = 0.167$
2W(C_{3v}): the same with $x = 0.346$
2W(C_{3v}): the same with $x = 0.448$

$D8_6$ ($Cu_{15}Si_4$ type)

Cubic: T_d^6–$I\bar{4}3d$; $a = 9.615$; $A = 76$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 12Cu($S_4$): $0\frac{1}{4}\frac{3}{8}$; ♪; $0\frac{3}{4}\frac{1}{8}$; ♪;
+48Cu(C_1): xyz; ♪; $x\bar{y}(\frac{1}{2}-z)$; ♪; $(\frac{1}{2}-x)y\bar{z}$; ♪; $\bar{x}(\frac{1}{2}-y)z$; ♪; $(\frac{1}{4}+y)(\frac{1}{4}+x)$
$(\frac{1}{4}+z)$; ♪; $(\frac{1}{4}-y)(\frac{1}{4}+x)(\frac{3}{4}-z)$; ♪; $(\frac{1}{4}+y)(\frac{3}{4}-x)(\frac{1}{4}-z)$; ♪; $(\frac{3}{4}-y)(\frac{1}{4}-x)$; $(\frac{1}{4}+z)$; ♪;
$x = 0.12$; $y = 0.16$; $z = 0.04$
+16Si(C_3): $xxx; x\bar{x}(\frac{1}{2}-x)$; ♪; $(\frac{1}{4}+x)(\frac{1}{4}+x)(\frac{1}{4}+x)$; $(\frac{1}{4}-x)(\frac{1}{4}+x)(\frac{3}{4}-x)$; ♪; $x = 0.208$

$D8_8$ (Mn_5Si_3 type)

Hexagonal: D_{6h}^3–$P6_3/mcm$; $a = 6.912(5), c = 4.812(5)$; $A = 16$
Co-ordinates: 4Mn(D_3): $\frac{1}{3}\frac{2}{3}0$; $\frac{2}{3}\frac{1}{3}0$; $\frac{1}{3}\frac{2}{3}\frac{1}{2}$; $\frac{2}{3}\frac{1}{3}\frac{1}{2}$
6Mn(C_{2v}): $\pm(x0\frac{1}{4}; 0x\frac{1}{4}; \bar{x}\bar{x}\frac{1}{4})$; $x = 0.23$
6Si(C_{2v}): the same with $x = 0.60$
For Mg_5Hg_3: $x(Mg) = 0.25$; $x(Hg) = 0.615$

$D8_9$ (Co_9S_8 type)

Cubic: O_h^5–$Fm3m$; $a = 9.930(2)$; $A = 68$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♪) + 4Co(O_h): $\frac{1}{2}\frac{1}{2}\frac{1}{2}$
+32Co(C_{3v}): $\pm(xxx; x\bar{x}\bar{x}; ♪)$; $x \approx \frac{1}{8}$
+8S(T_d): $\pm(\frac{1}{4}\frac{1}{4}\frac{1}{4})$
+24S(C_{4v}): $\pm(x00; ♪)$; $x \approx \frac{1}{4}$

$D8_{10}$ (Cr_5Al_8 type)

Rhombohedral: C_{3v}^5–$R3m$; $a = 7.79$; $\alpha = 109°8'$; $A = 26$
(Hexagonal setting: $a = 12.70, c = 7.90$; $A = 78$)
Co-ordinates: 1Cr(C_{3v}): xxx; $x = 0.097$
3Cr(C_s): xxz; ♪; $x = -0.103$; $z = 0.106$
3CR(C_s): the same with $x = 0.170$; $z = -0.172$
3Cr(C_s): the same with $x = 0.003$; $z = 0.352$
1Al(C_{3v}): xxx; $x = -0.164$
3Al(C_s): xxz; ♪; $x = 0.006$; $z = -0.352$
3Al(C_s): the same with $x = 0.291$; $z = 0.058$
3Al(C_3): the same with $x = -0.322$; $z = 0.044$
6Al(C_1): xyz; ♪; xzy; ♪; $x = 0.330$; $y = -0.297$; $z = -0.042$

$D8_{11}$ (Co_2Al_5 type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 7.656\,0, c = 7.593\,2$; $A = 28$
Co-ordinates: 2Co(D_{3h}): $\frac{2}{3}\frac{1}{3}\frac{1}{4}$; $\frac{1}{3}\frac{2}{3}\frac{3}{4}$
6Co(C_{2v}): $\pm(x, 2x, \frac{1}{4}; 2\bar{x}, \bar{x}, \frac{1}{4}; x\bar{x}\frac{1}{4})$; $x = 0.128$
2Al(D_{3d}): $000; 00\frac{1}{2}$
6Al(C_{2v}): $\pm(x, 2x, \frac{1}{4}; 2\bar{x}, \bar{x}, \frac{1}{4}; x\bar{x}\frac{1}{4})$; $x = 0.467$
12Al(C_s): $\pm(2x, x, z; \bar{x}, 2\bar{x}, z; \bar{x}xz; 2\bar{x}, \bar{x}, \frac{1}{2}+z; x, 2x, \frac{1}{2}+z; x, \bar{x}, \frac{1}{2}+z)$; $x = 0.196$;
$z = 0.061$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D8_a$ (Th_6MN_{23} type)

Cubic: O_h^5–$Fm3m$; $a = 12.443$; $A = 116$

Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♪) +24Th(C_{4v}): ±($x00$; ♪); $x = 0.207$
+4Mn(O_h): $\frac{1}{2}\frac{1}{2}\frac{1}{2}$
+24Mn(D_{2h}): $\frac{1}{4}\frac{1}{4}0$;♪; $\frac{1}{4}\frac{3}{4}0$; ♪
+32Mn(C_{3v}): ±(xxx; $x\bar{x}\bar{x}$; ♪); $x = 0.378$
+32Mn(C_{3v}): the same with $x = 0.178$

$D8_b$ (σ type, as for example: V_3Ni_2)

(a) According to Bergman and Shoemaker[202]:

Tetragonal: D_{4h}^{14}–$P4_2/mnm$; for σ-FeCr: $a = 8.7968(5)$, $c = 4.5585(3)$; $A = 30$

Co-ordinates: 2 atoms $A(D_{2h})$: 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$
4 atoms $B(C_{2v})$: ±($xx0$); ±($\frac{1}{2}+x, \frac{1}{2}-x, \frac{1}{2}$); $x = \frac{2}{5} = 0.400$
8 atoms $C(C_s)$: ±($xy0$); ±($yx0$); ±($\frac{1}{2}+x, \frac{1}{2}-y, \frac{1}{2}$); ±($\frac{1}{2}+y, \frac{1}{2}-x, \frac{1}{2}$);
$x = \frac{7}{15} = 0.468$; $y = \frac{2}{15} = 0.134$
8 atoms $D(C_s)$: the same with $x = \frac{11}{15} = 0.735$; $y = \frac{1}{15} = 0.067$
8 atoms $E(C_s)$: ±(xxz); ±($xx\bar{z}$); ±($\frac{1}{2}+x, \frac{1}{2}-x, \frac{1}{2}+z$); ±($\frac{1}{2}+x, \frac{1}{2}-x, \frac{1}{2}-z$);
$x = \frac{11}{16} = 0.183$; $z = \frac{1}{4} = 0.250$

The space groups C_v^4–$P4nm$ and D_{2d}^8–$P\bar{4}n2$ could not be ruled out.

(b) According to Kasper, Decker and Belanger,[203] who investigated σ-CoCr: the same space group; $a = 8.84(1)$, $c = 4.60(1)$; $A = 30$.

Co-ordinates: 2 atoms $A(D_{2h})$: $00\frac{1}{2}$; $\frac{1}{2}\frac{1}{2}0$
4 atoms B as under (a) but $x = 0.100$
8 atoms C as under (a) but $x = 0.373$; $y = 0.027$
8 atoms D as under (a) but $x = 0.573$; $y = 0.227$
8 atoms E as under (a) but $x = 0.300$; $z = \frac{1}{4}$

$D8_c$ ($Mg_2Cu_6Al_5$ type)

Cubic: T_h^1–$Pm3$; $a = 8.311(3)$; $A = 39$

Co-ordinates: 6Mg(C_{2v}): ±($x0\frac{1}{2}$; ♪); $x = 0.32$
6Cu(or Zn I) (C_{2v}): ♪ ±($x00$; ♪); $x = 0.225$
12Cu(or Zn II)(C_s): ±($\frac{1}{2}yz$; ♪); ±($\frac{1}{2}y\bar{z}$; ♪); $y = 0.243$; $z = 0.336$
1Al(or Zn III)(T_h): $\frac{1}{2}\frac{1}{2}\frac{1}{2}$
6Al(or Zn IV)(C_{2v}): ±($x\frac{1}{2}0$; ♪); $x = 0.16$
8Al(or Zn V) (C_3): ±(xxx; $x\bar{x}\bar{x}$♪); $x = 0.215$

For Mg_2Zn_{11}: x(Zn I) = 0.235; y(Zn II) = 0.243; z(Zn II) = 0.343; x(Zn IV) = 0.160; x(Zn V) = 0.222.

$D8_d$(Co_2Al_9)

Monoclinic: C_{2h}^5–$P2_1/c = 10.149(6)$, $b = 6.290(5)$, $c = 8.5565(5)$; $\beta = 142.40°$; $A = 22$

Co-ordinates: 2 Al I(C_i) = 000; $0\frac{1}{2}\frac{1}{2}$
4 times Al, II, Al III, Al IV, Al V and Co in: ±(yyz); ±($\frac{1}{2}+x, \frac{1}{2}-y, z$)

	Co	Al *II*	Al *III*	Al *IV*	Al *V*
x	0.3335	0.2682	0.2309	0.9986	0.0417
y	0.6149	0.9619	0.2899	0.1931	0.6148
z	0.2646	0.4044	0.0889	0.3891	0.2159

$D8_e$ ($Mg_{32}X_{49}$ type)

Cubic: T_h^5–$Im3$; $a = 14.16$; $A = 162$

Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 12MG($C_{2v}$): ±($x0\frac{1}{2}$; ♪); $x = 0.605$
+12 Mg(C_{2v}): ± the same with $x = 0.185$
+16 Mg(C_3): ±(xxx; $x\bar{x}\bar{x}$; ♪); $x = 0.185$
+24 Mg(C_s): ±($0yz$; ♪; $0y\bar{z}$; ♪); $y = 0.300$; $z = 0.115$
+2 Al(T_h): 000
+24 atoms (83% Zn; 17% Al)(C_s): ±($0yz$; ♪; $0y\bar{z}$; ♪); $y = 0.097$; $z = 0.157$
+24 atoms (44% Zn; 56% Al)(C_s): the same with $y = 0.195$; $z = 0.310$
+48 atoms (48% Zn; 52% Al)(C_s): ±(xyz; ♪; $xy\bar{z}$; ♪; $\bar{x}yz$;♪; $\bar{x}\bar{y}z$;♪);
$x = 0.160$; $y = 0.190$; $z = 0.400$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$D8_f$ (Ir_3Ge_7 type)

Cubic: O_h^9–$Im3m$; $a = 8.735(1)$; $A = 40$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) + 12 Ir($C_{4v}$): $\pm(x00; \curvearrowleft)$; $x = 0.342$
+12 Ge(D_{2d}): $\pm(\frac{1}{4}0\frac{1}{2}, \curvearrowleft)$
+16 Ge(C_{3v}): $(xxx; x\bar{x}\bar{x}; \curvearrowleft)$; $x = 0.156$

$D8_g$ (Mg_5Ga_2 type)

Orthorhombic: D_{2h}^{26}–$Ibam$; $a = 13.708, b = 7.017, c = 6.020$; $A = 28$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +8Ga($C_s$): $\pm(xy0)$; $\pm(x\bar{y}\frac{1}{2})$; $x = 0.122; y = 0.262$
+8Mg(C_s): the same with $x = 0.080; y = 0.660$
+8Mg(C_2): $\pm(x0\frac{1}{4})$; $\pm(\bar{x}0\frac{1}{4})$; $x = 0.242$
+4Mg(D_2): $\pm(00\frac{1}{4})$

[Unpublished work by E, Hellner; for Mg_5Tl_2: $x(Tl) \approx \frac{1}{8}$; $y(Tl) \approx \frac{1}{4}$]

$D8_h$ (W_2B_5 type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 2.98, c = 13.87$; $A = 14$
Co-ordinates: 4W(C_{3v}): $\pm(\frac{1}{3}\frac{2}{3}z)$; $\pm(\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z)$; $z = 0.139$
2B(D_{3h}): $\pm(00\frac{1}{4})$; +2B(D_{3h}): $\pm(\frac{1}{3}\frac{2}{3}\frac{3}{4})$
2B(D_{3d}): $000; 00\frac{1}{2}$
4B(C_{3v}): $\pm(\frac{1}{3}\frac{2}{3}z)$; $\pm(\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z)$; $z = -0.028$

$D8_i$ (Mo_2B_5 type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$; $a = 7.19$; $\alpha = 24°10'$; $A = 7$
(Hexagonal setting: $a = 3.01, c = 20.93$; $A = 21$)
Co-ordinates (for hex. setting): (000; $\frac{1}{3}\frac{2}{3}\frac{1}{3}$; $\frac{2}{3}\frac{1}{3}\frac{2}{3}$) +6Mo($C_{3v}$): $\pm(00z)$; $z = 0.075$
+6B(C_{3v}): the same with $z = \frac{1}{3}$
+6B(C_{3v}): the same with $z = 0.186$
+3B(C_{3d}): $00\frac{1}{2}$

$D8_k$ (Th_7S_{12} type)

Hexagonal: C_{6h}^5–$P6_3/m$; $a = 11.04, c = 3.98$; $A = 19$
Co-ordinates: 1Th(S_6): $\pm(00\frac{1}{4})$
6Th(C_s): $\pm(xy\frac{1}{4})$; $\pm(\bar{y}, x - y, \frac{1}{4})$; $\pm(y - x, \bar{x}, \frac{1}{4})$; $x = 0.153; y = -0.283$
6S(C_s): the same with $x = 0.514; y = 0.375$
6S(C_s): the same with $x = 0.235; y = 0 \pm 0.010$

$E1_1$ ($CuFeS_2$ type)

Tetragonal: D_{2d}^{12}–$I\bar{4}2d$; $a = 5.2864(8), c = 10.4102(8)$; $A = 16$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$ +4*Cu*(S_4): $000; \frac{1}{2}0\frac{1}{4}$;
+4*Fe*(S_4): $00\frac{1}{2}; \frac{1}{2}0\frac{3}{4}$
+8*S*(C_2): $\frac{1}{4}y\frac{1}{8}; \frac{3}{4}\bar{y}\frac{1}{8}; y\frac{37}{48}; \bar{y}\frac{17}{48}; y = 0.27$

$E1_a$ ($MgCuAl_2$ type)

Orthorhombic: D_{2h}^{17}–$Cmcm$; $a = 4.00, b = 9.23, c = 7.14$; $A = 16$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$) +4Mg(C_{2v}): $\pm(0y\frac{1}{4})$; $y = 0.072$
+4Cu(C_{2v}): $\pm(0y\frac{1}{4})$; $y = -0.222$
+8Al(C_s): $\pm(0yz)$; $\pm(0, y, \frac{1}{2} - z)$; $y = 0.356; z = 0.056$

$E1_b$ [$AuAgTe_4$ (Sylvanite) type]

Monoclinic: C_{2h}^4–$P2/c$; $a = 8.96(2), b = 4.49(2), c = 14.62(2)$; $\beta = 145°26'$; $A = 12$
Co-ordinates: 2Au(C_i): $000; 00\frac{1}{2}$
2Ag(C_2): $\pm(0y\frac{1}{4}; y = 0.433$
4Te(C_1): $\pm(xyz)$; $\pm(x, \bar{y}, \frac{1}{2} + z)$; $x = 0.298; y = 0.031; z = 0.999$
4Te(C_1): the same with $x = 0.277; y = 0.425; z = 0.235$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$E9_3$ (Fe_3W_3C type)

Cubic: O_h^7–$Fd3m$; $a = 11.10$; $A = 112$

Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♪) +16Fe(D_{3d}): $\frac{5}{8}\frac{5}{8}\frac{5}{8}$; $\frac{5}{8}\frac{7}{8}\frac{7}{8}$; ♪
+32Fe(C_{3v}): $xxx; x\bar{x}\bar{x}$; ♪; $(\frac{1}{4}-x)(\frac{1}{4}-x)(\frac{1}{4}-x)$; $(\frac{1}{4}-x)(\frac{1}{4}+x)(\frac{1}{4}+x)$; ♪; $x = 1.175$
+48W(C_{2v}): $\pm(x00;$ ♪); $(\frac{1}{4}+x)\frac{1}{4}\frac{1}{4}$; ♪; $(\frac{1}{4}-x)\frac{1}{4}\frac{1}{4}$; $x = 0.195$
+16C(C_{3d}): $\frac{1}{8}\frac{1}{8}\frac{1}{8}$; $\frac{1}{8}\frac{3}{8}\frac{3}{8}$; ♪

$E9_4$ (Al_5C_3N type)

Hexagonal: C_{6v}^4–$P6_3mc$; $a = 3.285$; $c = 21.65$; $A = 18$

Co-ordinates: 2Al(C_{3v}): $00z$; $00(\frac{1}{2}+z)$; $z = 0.150$
2Al(C_{3v}): the same with $z = 0.345$
2Al(C_{3v}): $\frac{1}{3}\frac{2}{3}z$; $\frac{2}{3}\frac{1}{3}(\frac{1}{2}+z)$; $z = 0.045$
2Al(C_{3v}): the same with $z = 0.456$
2Al(C_{3v}): the same with $z = 0.240$
2C(C_{3v}) the same with $z = 0.133$
2C(C_{3v}): the same with $z = 0.369$
2C(C_{3v}): $00z$; $00(\frac{1}{2}+z)$; $z = 0.001$
2N(C_{3v}): the same with $z = 0.250$

$E9_a$ ($FeCu_2Al_7$ type)

Tetragonal: D_{4h}^6–$P4/mnc$; $a = 6.336(1)$, $c = 14.870(2)$; $A = 40$

Co-ordinates: 4Fe(C_4): $\pm(00z)$; $\pm(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}+z)$; $z = 0.300$
8Cu(C_s): $\pm(xy0)$; $\pm(\frac{1}{2}+x, \frac{1}{2}-y, \frac{1}{2})$; $\pm(\bar{y}x0)$; $\pm(\frac{1}{2}+y, \frac{1}{2}+x, \frac{1}{2})$; $x = 0.278$; $y = 0.092$
4Al(C_4): $\pm(00z)$; $\pm(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}+z)$; $z = 0.122$
8Al(C_2): $\pm(x, \frac{1}{2}+x, \frac{1}{4}; \bar{x}, \frac{1}{2}-x, \frac{1}{4}; \frac{1}{2}-x, x, \frac{1}{4})$; $x = 0.167$
16Al(C_1): $\pm(xyz; \bar{x}\bar{y}z; \bar{y}xz; y\bar{x}z; \frac{1}{2}+x, \frac{1}{2}-y, \frac{1}{2}+z; \frac{1}{2}-x, \frac{1}{2}+y, \frac{1}{2}+z;$
$\frac{1}{2}+y, \frac{1}{2}+x, \frac{1}{2}+z; \frac{1}{2}-y, \frac{1}{2}-x, \frac{1}{2}+z)$; $x = 0.203$; $y = 0.414$; $z = 0.100$

$E9_b$ ($FeMg_3Al_8Si_6$ type)

Hexagonal: D_{3h}^3–$P\bar{6}2m$; $a = 6.63$, $c = 7.94$; $A = 18$

Co-ordinates: 1Fe(D_{3h}): 000
3Mg(C_{2v}): $x0\frac{1}{2}$; $0x\frac{1}{2}$; $\bar{x}\bar{x}\frac{1}{2}$; $x = 0.445$
1Al(D_{3h}): $00\frac{1}{2}$
3Al(C_{2v}): $x00$; $0x0$; $\bar{x}\bar{x}0$; $x = 0.403$
4Al(C_3): $\pm(\frac{1}{3}\frac{2}{3}z)$; $\pm(\frac{1}{3}\frac{2}{3}\bar{z})$; $z = 0.231$
6Si(C_s): $x0z; x0\bar{z}; 0xz; 0x\bar{z}; \bar{x}\bar{x}z; \bar{x}\bar{x}\bar{z}$; $x = 0.750$; $z = 0.223$

$E9_c$ (Mn_3Al_9Si type)

Hexagonal: D_{4h}^6–$P6_3/mmc$; $a = 7.42$, $c = 7.72$; $A = 26$

Co-ordinates: 6Mn(C_{2v}): $\pm(x, 2x, \frac{1}{4})$; $\pm(2x, x, \frac{3}{4})$; $\pm(x\bar{x}\frac{1}{4})$; $x = 0.120$
6Al(C_{2v}): the same with $x = 0.458$
12Al(C_s): $\pm(x, 2x, z; 2x, x, \bar{z}; x\bar{x}z; x, 2x, \frac{1}{2}-z; 2x, x, \frac{1}{2}+z; x, \bar{x}, \frac{1}{2}-z)$; $x = 0.201$;
$z = -0.067$
2Si(D_{3d}): 000; $00\frac{1}{2}$

$E9_d$ ($AlLi_3N_2$ type)

Cubic: T_h^7–$Ia3$; $a = 9.48$; $A = 96$

Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +16Al($C_3$): $\pm(xxx)$; $\pm(\frac{1}{2}+x, \frac{1}{2}-x, \bar{x}$; ♪); $x = 0.115$
+48Li(C_1): $\pm(xyz;$ ♪); $\pm(x, \bar{y}, \frac{1}{2}-z;$ ♪); $\pm(\frac{1}{2}-x, y, \bar{z};$ ♪);
$\pm(\bar{x}, \frac{1}{2}-y, z;$ ♪); $x = 0.160$; $y = 0.382$; $z = 0.110$
+8N(C_{3i}): 000; $\frac{1}{2}\frac{1}{2}0$; ♪
+24N(C_2): $\pm(x0\frac{1}{4};$ ♪); $\pm(\bar{x}\frac{1}{2}\frac{1}{4};$ ♪); $x = 0.205$

For $GaLi_3N_2$: x(Ga) = 0.117; x(Li) = 0.152; y(Li) = 0.381; z(Li) = 0.114; x(N) = 0.215

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$E9_e$[$CuFe_2S_3$ (Cubanite) type]

Orthorhombic: D_{2h}^{16}–*Pnma*; $a = 6.231(2), b = 11.117(6), c = 6.467(1); A = 24$
Co-ordinates: 4Cu(C_s): $\pm(x\,\frac{1}{4}\,z)$; $\pm(\frac{1}{2}-x, \frac{3}{4}, \frac{1}{2}+z); x = \frac{1}{8}; z = \frac{7}{12}$
8Fe(C_1): $\pm(xyz; \frac{1}{2}+x, \frac{1}{2}-y, \frac{1}{2}-z; \bar{x}, \frac{1}{2}+y, \bar{z}; \frac{1}{2}-x, \bar{y}, \frac{1}{2}+z); x = \frac{1}{8}; y = \frac{1}{12}; z = \frac{1}{12}$
8S(C_1): the same with $x = \frac{1}{4}; y = \frac{1}{12}; z = \frac{5}{12}$
4S(C_s): $\pm(x\,\frac{1}{4}\,z)$; $\pm(\frac{1}{2}-x, \frac{3}{4}, \frac{1}{2}+z); x = \frac{1}{4}; z = \frac{11}{12}$

$F0_1$ [NiSbS (Ullmannite) type]

Cubic: T^4–$P2_13$; $a = 5.933(3); A = 12$
Co-ordinates: 4Ni(C_3): $xxx; (\frac{1}{2}+x)(\frac{1}{2}-x)\bar{x}$; ♪; $x \approx 0$
4Sb(C_3): the same with $x \approx 0.385$
4Sb(C_3): the same with $x \approx 0.615$

$F5_1$ ($NaHF_2$ type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$; $a = 5.05; \alpha = 40°2'; A = 4$
(Hexagonal setting: $a = 3.45, c = 13.90, c/a = 4.03; A = 12$)
Co-ordinates: 1Na(D_{3d}): 000; 1H(D_{3d}) = $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}$
2F(C_{3v}): $\pm(xxx); x = 0.410$
For $CaCN_2$: $\alpha = 43°50'; c/a = 3.63; x = 0.37$
For $NaCrS_2$: $\alpha = 29°48'; c/a = 5.59; x = 0.236$
For $NaCrSe_2$: $\alpha = 30°18'; c/a = 5.49; x = 0.235$
For $RbCrSe_2$: $\alpha = 21°33'; c/a = 7.85$

$F5_6$ ($CuSbS_2$ type)

Orthorhombic: D_{2h}^{16}–*Pnma*; $a = 6.008(10), b = 3.784(10), c = 14.456(30); A = 16$
Co-ordinates: 4Cu(C_s): $\pm(x\,\frac{1}{4}\,z)$; $\pm(\frac{1}{2}+x, \frac{1}{4}, \frac{1}{2}-z); x = 0.25; z = 0.83$
4Sb(C_s): the same with $x = 0.23; z = 0.06$
4S(C_s): the same with $x = 0.63; z = 0.10$
4S(C_s): the same with $x = 0.88; z = 0.83$

$F5_a$ ($KFeS_2$ type)

Monoclinic: C_{2h}^6–C2/c; $a = 7.079(1), b = 11.304(2), c = 5.398(1); \beta = 113.20(1); A = 16$
Co-ordinates: (000; $\frac{1}{2}\,\frac{1}{2}\,0$) +4K($C_2$): $\pm(0y\,\frac{1}{4}); y = 0.355$
+Fe(C_2): the same with $y = -0.008$
+8S(C_1): $\pm(xyz); \pm(x, \bar{y}, \frac{1}{2}+z); x = 0.195; y = 0.111; z = 0.10$

$H1_1$ [Spinel (Al_2MgO_4) type]

Cubic: O_h^7–$Fd3m$; $a = 8.075; A = 56$
Co-ordinates: (000; $\frac{1}{2}\,\frac{1}{2}\,0$; ♪) +8Mg($I_d$): 000; $\frac{1}{4}\,\frac{1}{4}\,\frac{1}{4}$
+16Al(D_{3d}): $\frac{5}{8}\,\frac{5}{8}\,\frac{5}{8}; \frac{5}{8}\,\frac{7}{8}\,\frac{7}{8}$; ♪
+32O(C_{3v}): $xxx; x\bar{x}\bar{x}$; ♪; $(\frac{1}{4}-x)(\frac{1}{4}-x)(\frac{1}{4}-x); (\frac{1}{4}-x, \frac{1}{4}+x, \frac{1}{4}+x)$; ♪; $x \approx -\frac{1}{8}$

In some compounds, better agreement with observed intensities is obtained by assuming that the metal atoms are distributed at random among the 24 available sites, or that the trivalent element occupies all the 8-equivalent sites and half of the 16-equivalent ones. In some cases lattice sites may be vacant, e.g., γ-Al_2O_3 or In_2S_3.

$H2_4$[Cu_3VS_4 (Sulvanite) type]

Cubic: T_d^1–$P\bar{4}3m$; $a = 5.391(5); A = 8$
Co-ordinates: 3Cu(D_{2d}): $\frac{1}{2}00; 0\,\frac{1}{2}\,0; 00\,\frac{1}{2}$
1V(T_d): 000
4S(C_{3v}): $xxx; x\bar{x}\bar{x}$; ♪; $x = 0.235$

$H2_6$ [Stannite ($FeCu_2SnS_4$) type]

Tetragonal: D_{2d}^{11}–$I\bar{4}2m$; $a = 5.449(2); c = 10.757(3); A = 16$
Co-ordinates: (000; $\frac{1}{2}\,\frac{1}{2}\,\frac{1}{2}$) +2Fe($D_{2d}$): 000
+2Sn(D_{2d}): $00\,\frac{1}{2}$
+4Cu(S_4): $0\,\frac{1}{2}\,\frac{1}{4}; \frac{1}{2}\,0\,\frac{1}{4}$
+8S(C_s): $xxz; \bar{x}\bar{x}z; x\bar{x}\bar{z}; \bar{x}x\bar{z}; x = 0.245; z = 0.132$

(*continued*)

Table 6.2 STRUCTURAL DETAILS—*continued*

$L1_0$ (CuAu type)

Tetragonal: D_{4h}^1–$P4/mmm$; $a = 3.964$; $c = 3.672$; $A = 4$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$) +2Cu(D_{4h}): 000
+2Au(D_{4h}): $\frac{1}{2}0\frac{1}{2}$
Superstructure of the $A1$ (Cu) type

$L1_2$ (Cu_3Au type)

Cubic: O_h^1–$Pm3m$; $a = 3.747$; $A = 4$
Co-ordinates: 3Cu(D_{4h}): $\frac{1}{2}\frac{1}{2}0$; ♪
1Au(O_h): 000
Superstructure of $A1$ (Cu) type

$L2_1$ (Cu_2MnAl type)

Cubic: O_h^5–$Fm3m$; $a = 5.968$; $A = 16$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}0$; ♪ +4Al(O_h): 000
+8Cu(I_d): $\frac{1}{4}\frac{1}{4}\frac{1}{4}$; $\frac{3}{4}\frac{3}{4}\frac{3}{4}$
+4Mn(O_h): $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

Superstructure of the $A2$ (W) type; this type is virtually identical with the D_{o3} (BiF_3 or $BiLi_3$) type

$L2_2$ (Tl_7Sb_2 type)

Cubic: O_h^9–$Im3m$; $a = 11.618(6)$; $A = 54$
Co-ordinates: (000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$) +2Tl($O_h$): 000
+16Tl(C_{3v}): $\pm(xxx, x\bar{x}\bar{x}; ♪)$; $x = 0.17 \approx \frac{1}{6}$
+24Tl(C_{2v}): $\pm(xx0; ♪; x\bar{x}0; ♪)$; $x = 0.35 \approx \frac{1}{3}$
+12Sb(C_{4v}): $\pm(x00; ♪)$; $x = 0.29 \approx \frac{1}{3}$

L_1' (Fe_4N type)

Cubic: O_h^1–$Pm3m$ or O_h^5–$Fm3m$ (depending on the distribution of the N atoms); $a = 3.896(2)$; $A = 5$
Co-ordinates: 4Fe at 000; $\frac{1}{2}\frac{1}{2}0$; ♪
1N at $\frac{1}{4}\frac{1}{4}\frac{1}{4}$; or at $\frac{1}{2}\frac{1}{2}\frac{1}{2}$, probably the latter; or at random at $\frac{1}{4}\frac{1}{4}\frac{1}{4}$; $\frac{3}{4}\frac{3}{4}\frac{1}{4}$; ♪; and (or) at $\frac{3}{4}\frac{3}{4}\frac{3}{4}$; $\frac{1}{4}\frac{1}{4}\frac{3}{4}$; ♪; and (or) at $\frac{1}{2}\frac{1}{2}\frac{1}{2}$; $\frac{1}{2}00$; ♪

$L'1_2$ (~$AlFe_3C$ type)

Cubic: O_h^1–$Pm3m$; $a = 3.78$; $A \approx 5$
Co-ordinates: 1Al(O_h): 000
3Fe(D_{4h}): $\frac{1}{2}00$; ♪
0.6 to 0.9C(O_h) at $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

$L'2$ (Martensite type)

Tetragonal: D_{4h}^{17}–$I4/mmm$; $a = 2.84$, $c = 2.97$; $A =$ 2Fe + (up to) 0.12C
Co-ordinates: 2Fe(D_{4h}) at 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$
The C atoms at random: $\frac{1}{2}\frac{1}{2}0$ and (or) $00\frac{1}{2}$

$L'3$ (Interstitial $A3$ type)

Hexagonal: D_{6h}^4–$P6_3/mmc$ or D_{6h}^1–$P6/mmm$
Co-ordinates: 2 metal atoms (D_{3h}): $\frac{2}{3}\frac{1}{3}0$; $\frac{1}{3}\frac{2}{3}\frac{1}{2}$
C or N(C_{3v}): $\frac{1}{3}\frac{2}{3}z$; $\frac{2}{3}\frac{1}{3}(\frac{1}{2}+z)$; $\frac{1}{3}\frac{2}{3}\bar{z}$; $\frac{2}{3}\frac{1}{3}(\frac{1}{2}-z)$; $z \approx \frac{3}{8}$; or $00\frac{1}{4}$; $00\frac{3}{4}$

$L6_0$ (Ti_3Cu type)

Tetragonal: D_{4h}^1–$P4/mmm$; $a = 4.158$, $c = 3.594$; $A = 4$
Co-ordinates: 1Cu(D_{4h}): 000
1Ti(D_{4h}): $\frac{1}{2}\frac{1}{2}0$
2Ti(D_{2h}): $0\frac{1}{2}\frac{1}{2}$; $\frac{1}{2}0\frac{1}{2}$

Tetragonal deformed $L1_2$ (Cu_3Au) type; superstructure of $A6$ (In) type

$L'6$ (Interstitial $A6$ type)

Tetragonal: D_{4h}^{17}–$F4/mmm$ (or D_{4h}^1–$P4/mmm$, depending on the distribution of the N atoms)
Co-ordinates: 4 metal atoms (D_{4h}): 000; $\frac{1}{2}\frac{1}{2}0$; ♪;
N atoms in the holes: $\frac{1}{4}\frac{1}{4}\frac{1}{4}$; $\frac{1}{4}\frac{3}{4}\frac{3}{4}$; ♪; $\frac{3}{4}\frac{3}{4}\frac{3}{4}$; $\frac{3}{4}\frac{1}{4}\frac{1}{4}$; ♪; or $\frac{1}{2}\frac{1}{2}\frac{1}{2}$; $\frac{1}{2}00$; ♪

Table 6.3 COMPARISON OF STRUKTURBERICHT AND PEARSON NOMENCLATURE

Strukturbericht structural type	*Typical compound or element*	*Pearson symbol*
*A*1	Cu	cF4
*A*2	W	cI2
*A*3	Mg	hP2
*A*4	C (Diamond)	cF8
*A*5	Sn (Beta)	tI4
*A*6	In	tI2
*A*7	As	hR2
*A*8	Gamma-Se	hP3
*A*9	C (Graphite)	hP4
*A*10	Hg	hR1
*A*11	Ga	oC8
*A*12	Alpha-Mn	cI58
*A*13	Beta-Mn	cP20
*A*15	Cr_3Si	cP8
*A*16	Alpha-S	oF128
*A*20	Alpha-U	oC4
A_a	Pa	tI2
A_b	Beta-U	tP30
A_c	Alpha-Np	oP8
A_d	Beta-Np	tP4
A_e	Beta-$TiCu_3$	oF4
A_f	$HgSn_{10}$	hP1
A_g	B	tP50
A_h	Alpha-Po	cP1
A_i	Beta-Po	hR1
A_k	Alpha-Se	mP32
A_l	Beta-Se	mP32
*B*1	NaCl	cF8
*B*2	CsCl	cP2
*B*3	ZnS (Sphalerite)	cP2
*B*4	ZnS (Würtzite)	hP4
$B8_1$	NiAs	hP4
$B8_2$	Ni_2In	hP6
*B*9	HgS	hP6
*B*10	LiOH; PbO	tp4
*B*13	NiS	hR6
*B*16	GeS	oP8
*B*17	PtS	tP4
*B*18	CuS	hP12
*B*19	AuCd	oP4
*B*20	FeSi	cP8
*B*27	FeB	oP8
*B*29	SnS	oP8
*B*31	MnP	oP8
*B*32	NaTl	cF16
*B*34	PdS	tP16
*B*35	CoSn	hP6
*B*37	TlSe	tI16
B_a	UCo	cI16
B_b	AgZn	hP9
B_c	CaSi	oP16
B_d	NiSi	oP8
B_e	CdSb	oP16
B_f	CrB	oC8
B_g	MoB	tI16
B_h	WC	hP2
B_i	MoC	hP8
B_l	AsS	mP32
B_m	TiB	oP8
*C*1	CaF_2	cF12
*C*2	FeS_2	cP12
*C*6	CdI_2	hP3
*C*7	MoS_2	hP6

(*continued*)

Table 6.3 COMPARISON OF STRUKTURBERICHT AND PEARSON NOMENCLATURE—*continued*

Strukturbericht structural type	*Typical compound or element*	*Pearson symbol*
$C11_a$	CaC_2	tI6
$C11_b$	$MoSi_2$	tI6
$C12$	$CaSi_2$	hR6
$C14$	$MgZn_2$	hP12
$C15$	$MgCu_2$	cF24
$C15_b$	$AuBe_5$	cF24
$C16$	$CuAl_2$	tI12
$C18$	FeS_2 (Marcasite)	oP6
$C22$	Fe_2P	hP9
$C23$	$PbCl_2$	oP12
$C32$	AlB_2	hP3
$C33$	Bi_2Te_2S	hR5
$C34$	$AuTe_2$ (Calaverite)	mC6
$C36$	$MgNi_2$	hP24
$C37$	Co_2Si	oP12
$C38$	Cu_2Sb	tP6
$C40$	$CrSi_2$	hP9
$C42$	SiS_2	oI12
$C44$	GeS_2	oF72
$C46$	$AuTe_2$ (Krennerite)	oP24
$C49$	$ZrSi_2$	oC12
$C54$	$TiSi_2$	oF24
C_a	Mg_2Ni	hP18
C_b	Mg_2Cu	oF48
C_c	$ThSi_2$	tI12
C_e	$CoGe_2$	oC24
C_g	ThC_2	mC12
C_h	$CuTe_2$	hP6
$D0_2$	$CoAs_3$	cI32
$D0_3$	BiF_3	cF16
$D0_9$	ReO_3/Cu_3N	cF4
$D0_{11}$	Fe_3C	oP16
$D0_{18}$	Na_3As	hP8
$D0_{19}$	Mg_3Cd	hP8
$D0_{20}$	$NiAl_3$	oP16
$D0_{21}$	Cu_3P	hP24
$D0_{22}$	$TiAl_3$	oP16
$D0_{23}$	$ZrAl_3$	tI16
$D0_{24}$	$TiNi_3$	hP16
$D0_a$	$TiCu_3$	oP8
$D0_c$	U_3Si	tI16
$D0_d$	Mn_3As	oP16
$D1_3$	$BaAl_4$	tI10
$D1_a$	$MoNi_4$	tI10
$D1_b$	UAl_4	oI20
$D1_c$	$PtSn_4$	oC20
$D1_d$	$PtPb_4$	tP10
$D1_e$	UB_4	tP20
$D1_f$	MnB_4	oF40
$D1_g$	B_4C	hR15
$D2_1$	CaB_6	cP7
$D2_3$	$NaZn_{13}$	cF112
$D2_b$	$ThMn_{12}$	tI26
$D2_c$	U_6Mn	tI26
$D2_d$	$CaCu_5$	hP6
$D2_e$	$BaHg_{11}$	cP36
$D2_f$	UBe_{12}	cF52
$D2_g$	Fe_8N	tI18
$D5_2$	La_2O_3	hP5
$D5_3$	Mn_2O_3	cI80
$D5_8$	Sb_2S_3	oP20
$D5_9$	Zn_3P_2	tP40

(*continued*)

Table 6.3 COMPARISON OF STRUKTURBERICHT AND PEARSON NOMENCLATURE—*continued*

Strukturbericht structural type	*Typical compound or element*	*Pearson symbol*
$D5_{10}$	Cr_3C_2	oP20
$D5_{13}$	Ni_2Al_3	hP5
$D5_a$	U_3Si_2	tP10
$D5_b$	Pt_2Sn_3	hP10
$D5_c$	Pu_2C_3	cI40
$D5_e$	Ni_3S_2	hR5
$D5_f$	As_2S_3	mP20
$D7_1$	Al_4C_3	hP7
$D7_2$	Co_3S_4	cF56
$D7_3$	Th_3P_4	cI28
$D7_a$	Ni_3Sn_4	mC14
$D7_b$	Ta_3B_4	oI14
$D8_1$	Fe_3Zn_{10}	cI52
$D8_2$	Cu_5Zn_8	cI52
$D8_3$	Cu_9Al_4	cP52
$D8_4$	$Cr_{23}C_6$	cF116
$D8_5$	Fe_7W_6	hR13
$D8_6$	$Cu_{15}Si_4$	cI76
$D8_8$	Mn_5Si_3	hP16
$D8_9$	Co_9S_8	cF68
$D8_{10}$	Cr_5Al_8	hR26
$D8_{11}$	Co_2Al_5	hP28
$D8_a$	Th_6Mn_{23}	cF116
$D8_b$	Sigma (FeCr)	tP30
$D8_c$	$Mg_2Cu_6Al_5$ (Mg_2Zn_{11})	cP39
$D8_d$	Co_2Al_9	mP22
$D8_e$	$Mg_{32}X_{49}$ (X = Al, Zn)	cI162
$D8_f$	Ir_3Ge_7	cI40
$D8_g$	Mg_5Ga_2	oI28
$D8_h$	W_2B_5	hP14
$D8_i$	Mo_2B_5	hR7
$D8_k$	Th_7S_{12}	hP19
$E1_1$	$CuFeS_2$	tI16
$E1_a$	$MgCuAl_2$	oC16
$E1_b$	$AuAgTe_4$	mP12
$E9_3$	Fe_3W_3C	cF112
$E9_4$	Al_5C_3N	hP18
$E9_a$	$FeCu_2Al_7$	tP40
$E9_b$	$FeMg_3Al_8Si_6$	hP18
$E9_c$	Mn_3Al_9Si	hP26
$E9_d$	$AlLi_3N_2$	cI96
$E9_e$	$CuFe_2S_3$	oP24
$F0_1$	NiSbS	cP12
$F5_1$	$NaHF_2$	hR4
$F5_6$	$CuSbS_2$	oP16
$F5_a$	$KFeS_2$	mC16
$H1_1$	$MgAl_2O_4$	cF56
$H2_4$	Cu_3VS_4	cP8
$H2_6$	$FeCu_2SnS_4$	tP16
$L1_0$	CuAu	tP2
$L1_2$	Cu_3Au	cP4
$L2_1$	Cu_2MnAl	cF16
$L2_2$	Tl_7Sb_2	cI54
L'_1	Fe_4N	cF5
$L'1_2$	$AlFe_3C$	cP5
$L'2$	Fe-C (Martensite)	tI2
$L'3$	Fe_2N	hP3
$L6_0$	$CuTi_3$	tP4

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Supplemented by checking of selected phases not present in the above (NaCl, CsCl, LiOH, $PbCl_2$, P) using the Powder Diffraction File, International Center for Diffraction Data, Newton Square, PA, 2001.